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An Experimental Study of Mother Loss in the Indian Langur Monkey *(Presbytis entellus)*

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Key Words. Langur monkey · *Presbytis entellus* · Mother loss · Caregiving · Depression · Social behavior

Abstract. Langur monkeys normally have many important caregivers from birth. Mothers of 6- to 8-month-old infants were removed from the infants' social groups for 2 weeks, and the effects of the separations on infant behavior were assessed. All infants showed changes in behavior. Their reactions to loss ranged from minimal to severe – in 2 instances, death. All infants sought care during separation, and most infants showed strong preferences for certain females. A majority of infants adopted a major caregiver during the mother's absence. Adoptions were entirely infant-initiated, and most infants remained with their adopted caregivers when the mothers were returned. Factors influencing infant response to mother loss included previous quality of maternal care, availability of accepting caregivers, and peer competition.

Introduction

Clinical interest has focused on human situations in which mother loss is a major factor – a context wherein depression is often observed. Separation of the human mother and infant has been studied and described by many investigators [for example, Ainsworth, 1962, 1969, 1974; Bowlby, 1951, 1960, 1969; Spitz, 1945, 1946; Yarrow, 1964]. Although the long-term effects of separation are not clear, the short-term effects have been well documented. Since it is impossible to study the results of separation and depression using humans as experimental subjects, researchers have sought to use nonhuman primates as research subjects and models.

In the wild, an infant Old World monkey or ape which has lost its mother faces an immediate threat to its existence. During the period of time between

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mother loss and the acquisition of substitute maternal care, if such is available, the infant is in grave jeopardy. The model of infant development common to the well-known Cercopithecines, including baboons and macaques, views the mother as primary if not exclusive caregiver. For these monkeys, loss of the mother in the wild means almost certain death to a very young infant. Even in captivity, the impact of mother loss on a pigtail macaque, for example, is such that severe depression, and in several instances death, has followed the experimental removal of the mother from an infant's social group.

The mother is the infant's source of nourishment, protection, and transportation. She is also the source of much information and serves as a model from which the infant learns many practical skills and many necessary facts about its physical and social environment [Hinde, 1974; Kaufman, 1970]. It appears that the phylogenetically closer the nonhuman primate is to humans, the more it has to acquire in the way of skills and knowledge. Hall [1963] demonstrated the importance of social learning and emphasized the mother-infant relationship as a context for early learning. He described the 'intimate relationship with positive emotional attitudes ... in the affectional context provided by the mother and others.' The mother ape or monkey is not only a necessary provider of physical comfort and safety but is one with and to which the infant quickly forms strong emotional ties. This is true even for the Colobines, which experience many caregivers from the first day of life. The mother-infant bond is the strongest and most persistent in the life of many Old World monkeys and apes; it may last many years [Hrdy, 1976; Kaufman, 1970; Sade, 1965, 1967; van Lawick-Goodall, 1965]. If the mother is not available, the functions she performs for the infant must be performed by some other individual if the young infant is to survive.

Several major laboratories have devoted years to the study of infant monkey reaction to loss of mother, and the results have pictured a depressive response of varying intensity after the loss [Abrams, 1969; Hansen, 1966; Hinde, 1974; Harlow and Suomi, 1974; Harlow et al. 1970; Jensen and Tolman, 1962; Kaplan, 1970, 1972; Kaufman, 1977; Kaufman and Rosenblum, 1967a, b, 1969; Mitchell et al., 1967; Preston et al., 1970; Reite et al., 1974; Rosenblum and Kaufman, 1967; Seay et al., 1962; Seay and Harlow, 1965; Spencer-Booth and Hinde, 1967, 1971]. Because the infant macaque's (*Macaca* spp.) response to separation is similar to the human depressive syndrome, macaques have been used in laboratory research as models for the study of human depression. Investigations of macaques have included behavioral, biochemical, and physiological studies [McKinney, 1974; McKinney and Bunney, 1969; McKinney et al..

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1971; Reite et al., 1974, 1978a–c]. Several major apects of social life have been, and are being, investigated as they relate to the results of experimental manipulations of, for example, peer attachments [Erwin, 1974; Patterson et al., 1975] and the environment [Chappel and Meier, 1975]. Sex differences have been studied [Jensen et al., 1966; Rosenblum, 1974a, b] and depressive behavior evaluated [Harlow and Suomi, 1974; Senay, 1973; Suomi et al., 1975; for a selected bibliography of early rearing of nonhuman primates with reduced conspecific contacts, see Roy, 1976, 1977]. Developmental sequences for different species of macaques are very similar; there is agreement from both field and laboratory work concerning the timing and sequence of stages [Hansen, 1966; Harlow, 1963; Hinde and Spencer-Booth, 1967; Imanishi, 1963; Kaufman, 1977; Rosenblum and Kaufman, 1967; Simonds, 1965].

Although there have been many field and laboratory investigations of primate behavior since the early 1960s, surprisingly few have been directed to the subfamily of Old World monkeys composed of the leaf-eating monkeys (Colobinae), which includes the common Indian languar monkey. The Colobinae comprise a large number of well-defined genera and species [Dolhinow, 1972; Napier and Napier, 1967; Simpson, 1945]. The great majority of investigations on the behavior of Old World nonhuman primates – including field studies, experimental manipulations, and psychological testing – have concentrated on the subfamily Cercopithecinae, especially the genera *Macaca* and *Papio.* This bias in our knowledge is even more fundamental than it appears because leaf-eating monkeys are rarely used in laboratories, and those studies which have been carried out have been exclusively of an anatomical or physiological nature [Amerasinghe et al., 1971; Ayer, 1948; Bauchop and Martucci, 1968; David and Ramaswami, 1969, 1971].

With its wide distribution and diverse habitats, the common Indian langur is the most variable of the Asian langurs. The genus *Presbytis* is a very widespread group of monkeys, and the species *Entellus* is the most wide-ranging of all Asian colobines. Its distribution extends through most of India and to parts of Kashmir, Nepal, Sikkim, Bangladesh and Sri Lanka. Langur troops inhabit a variety of ecological zones, from tropical, moist, deciduous forests to dry scrub with infrequent tall trees, as well as towns and villages where most of their diet is obtained from human cultivation. The social behavior of gray langur monkeys *(P. entellus)* has been relatively well studied under free-ranging field conditions [Bishop, 1975; Boggess, 1976; Curtin, 1975; Dolhinow, 1972; Hrdy, 1974, 1976, 1977; Jay, 1962, 1963, 1965; Mohnot, 1968, 1971a–c; Ripley, 1965, 1967, 1970; Sugiyama, 1964, 1965a, b, 1966, 1967; Sugiyama et al., 1965;

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Vogel, 1970, 1973, 1976; Weber, 1973; Yoshiba, 1967, 1968; see also review in Roonwall and Mohnot, 1976].

Field and laboratory studies indicate that langurs offer an illuminating contrast to the macaque model of infant reaction to mother loss. The langur monkey infant has many caregivers from the day of its birth, and it can be away from its mother for long periods of each day. Infant transfers, or passing young infants from one older animal to another, often continue until the infant is several months old. Langur infants are exposed to and handled by most if not all of the adult females in the group as well as by many immature females and some males, both mature and immature. Similar patterns of early experience are reported from other Colobines, both African and Asian.

The experimental research reported here has concentrated specifically on the effects of mother loss on langur infants. Mothers of 12 6- to 8-month-old infants were removed from the infants' social group for 2 weeks, and the effects of these separations on infant behavior were assessed. Results include the following.

(1) All infants showed changes in behavior during separation. Reactions ranged from minimal to very severe and, in two instances, death.

(2) All infants sought contact and care from other females during the separation, and most infants showed strong preferences for certain females. Most importantly, most of the infants adopted a female to serve as major and usually only caregiver during the mother's absence.

(3) All of these adoptions of substitute major caregivers were infant-initiated.

(4) Even after their mothers were returned to the group, most of the infants elected to remain with their adopted caregivers.

If we are to understand the complex and strong relationship between an infant and its mother or its major caregiver, we will have to investigate the behavior of the infant, the mother, and the social context in which the relationship is perpetuated. No one single line of investigation will allow understanding of this rich and vitally important social bond.

Materials and Methods

Animal Housing

The animals in this colony are housed in two cages, each a runway measuring 22.9 m in length, 6.0 m in width and 4.6 m in height. Each structure has a cement base, heavy wire cyclone mesh on the front and top, corrugated metal siding on the back and one end, and a

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Fig. 1. Each social group of langurs is housed in its own runway, a portion of which is seen in this photograph.

large heated sleeping room accessible at all times (fig. 1). Both segments of each cage are equipped with running water and have at least one partition with double-door entry way. Heavy branches, ropes, tires and other objects provide off-the-ground sitting for individuals, and large perches provide surfaces above ground for resting groups.

Animals are fed in the morning immediately after the cages are cleaned, and water is available at all times.

Subjects

The animal subjects used in this research live in two social groups. The colony was established in March 1972, and in late 1978 the 32-member, three-generational colony was composed of 3 adult males, 9 adult females, 2 subadult males, 6 juvenile and older infant males, 1 juvenile female, 7 infant males, and 4 infant females. 21 members of the colony were born in the Berkeley colony and 7 at the California Primate Research Center, Davis, Calif. 28 of the 31 animals therefore have known birth dates.

Observation Procedures

Every animal in the colony is sampled 6 times (for a total of 60 min) per week to maintain baseline records. More than 5,000 h of recorded samples are available as background information on the members of the colony since 1972. Data are obtained by use of a

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focal animal method of sampling as described by Altmann [1974]. Each sample is centered on a specific animal, and all activities of and interactions with the focal animal are recorded. Time length of a sample is 10 min, and both the frequency and duration of behaviors occurring during the sample period are recorded. It is thus possible to assess not only the frequency, duration and interactional patterning of behavioral events but also the sequencing, simultaneity and potential for latency (pauses) between behaviors. Observations are scheduled at different times of the day, and samples taken on a given day are not done continuously.

The behavior repertoire designed for use in this and related research [Dolhinow, 1978] was developed after careful examination of descriptions of particular behaviors published by other investigators of the genus *Presbytis* [e.g., Poirier, 1970; Ripley, 1965] in order to arrive at mutually comparable units where appropriate. The behavior units were also selected on the basis of the author's observations of langurs in India and after extensive preliminary observations of this colony group. To assure comparability of our data with those of other studies of attachment and separation, the langur monkey repertoire was developed with specific attention to the behavior repertoire published by Kaufman and Rosenblum [1966] for bonnet and pigtail macaques.

Separations of mothers from infants were scheduled to avoid overlap; hence every animal was present during the separation period except for the mother of the subject infant. The mean age of the infants at the start of the experiment was 30 weeks. Each experimental separation was undertaken in three 2-week stages: preseparation, separation, and postseparation. During preseparation week 1, the infant was observed for 6 samples (totaling 60 min) daily and the mother for 3 (30 min). During preseparation week 2, the infant was observed for 12 samples (120 min) daily and the mother for 6 (60 min). The mother was then removed from the social group; during week 1 of separation, the infant was observed for 18 samples (180 min) each day, and in week 2 of separation for 12 samples daily (120 min; mothers were also observed during separation, and these results will be presented elsewhere). When the mother was returned, the infant was observed for 12 samples (120 min) daily for 1 week and the mother for 6 samples (60 min). During postseparation week 2, the infant was sampled 6 times daily (60 min) and the mother 3 (30 min). Sampling was done 7 days a week during the entire period of experimentation.

Increased numbers of focal animal samples were undertaken for 6 weeks on control subjects, also starting at 30 weeks of age. Each 10-min sample detailed the date, time of day, observer code, weather conditions, and proximity scores of the focal animal to all other animals in the social group. Proximity scores were made at the beginning and end of each sample. 10-min samples were divided into minute units for later retrieval of behavior durations.

Data Processing

Behavioral data are transferred from original code sheets to the computer disk file by an interactive data entry program. This program checks for inconsistencies and errors and allows for corrections before the data are sorted in a compact string form. Fortran programs parse the behavior strings and produce summary tables. Programs are written to scan samples for selected behavior sets, to tally the number of minutes in which the behaviors occurred at least once, to seek out bouts of selected behaviors within a set of samples, and

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to tally the number of bouts occurring. These programs give detailed printouts for each animal and describe the mode of the behavior (whether actor, solitary, recipient, or participant). Another program tallies the number of minutes an animal is observed in any of a set of selected behaviors and prints out the total durations for each of the animals involved irrespective of the behavior modes. Proximities of individuals are collected by another program that tabulates these data at the beginning and end of each sample. The parsing algorithm used in decoding the behavior strings is sufficiently general to be applied to other behavioral data in any vocabulary used.

Changes in behavior frequencies during the study were assessed with analyses of variance and Newman-Keuls tests. Nonparametric analysis (Friedman test) yielded identical results. Statistical computations were carried out with the BMD-P2V computer program.

Results

Infants subject to separation were between the ages of 6 and 8 months, with a mean age of 32 weeks at the beginning of separation. Each infant was observed before the preseparation period to determine whether it was eating sufficient solid foods to survive in the mother's absence. One naturally occurring separation occurred when a 4-month-old infant's mother died. It did not eat and did not survive.

Separations of 12 infants were undertaken, 8 males and 4 females. The sex ratio was determined by the sex of births in our colony. Of the 12 separations, 9 (7 males, 2 females) were completed, permitting us to observe these infants throughout all stages of separation. 2 infants (1 male, 1 female) died during separation, the male on day 2 and the female on day 3. Since our colony is housed out of doors, 1 other separation (male) had to be terminated on day 4 of separation because of freezing temperatures and snow. Data from observation of 5 control animals (3 males, 2 females) were compared with the results of observation of subjects that were separated. Data on control subjects will be presented in detail elsewhere and are not included here. Table I summarizes basic information for separation subjects and controls. This includes the identity of the infant, whether there are related animals in the group, the mother's parity and age, the infant's age at the beginning of separation, the caregiver during separation, and the outcome of the mother's absence.

Data are available for the occurrence (frequency or duration) of every behavior in the langur monkey repertoire [Dolhinow, 1978], but only those behaviors considered most diagnostic of the degree of reaction to loss of mother are presented here. These behaviors include vocalization, locomotion, play,

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Table 1. Data on the separated and control infants

Infant	Mother	Sibs and age	Mother's age years	Infant's age weeks	Length of separation days	Caregiver during separation	Outcome
Separated Infants							
AL (m)	LT	–	13+[1]	26	4	–	LT
AN (f)	MO	CR, 6 years	13+[1]	26	2	–	dead
JA (m)	SM	QU, 20 months	6½[1]	30	14	ST, BF	BF
BO (m)	GI	–	6½[2]	33	14	LT	LT
OZ (m)	HA	–	13+[1]	37	14	LT, LU	HA
SE (m)	WA	DD, 2 years RP, 3½ years LL, 8½ years	13+[1]	28	14	ST	WA
TA (f)	MO	CR, 7 years	13+[1]	35	14	LT	LT
PI (m)	GI	BO, 2 years	8[1]	33	14	–	GI
PH (m)	LL	XX, 2 years 8 months	9½[1]	29	14	–	LL
CO (f)	WA	LL, 10 years RP, 5 years DD, 3½ years	13+[1]	34	14	BF	BF
DA (m)	LU	–	8[1]	30	3	–	dead
RA (m)	RP	–	4½[3]	38	14	BF	BF
Control Infants							
SA (f)	SM	JA, 2 years QU, 3½ years	8[1]	32	–	–	SM
MI (f)	MO	TA, 2 years CR, 8 years	15+[1]	32	–	–	MO
VA (m)	GI	PI, 1¾ years	9[1]	32	–	–	GI
SI (m)	SM	JA, 3 years QU, 4½ years SA, 1½ years	9[1]	32	–	–	SM
SL (m)	RP	RA, 1½ years	6[1]	32	–	–	RP

[1] Multiparous.
[2] Primiparous.
[3] First live birth.

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grooming, approaches to and by other animals, contact, mouth, manipulate, eat, aggression given and received, and protection [see Dolhinow, 1978, for definitions of all behaviors used in this paper]. Detailed behavior graphs are presented below for typical individual infants. All figures are based on approximately equal observation sample sizes, and the same observational procedures were used in every separation.

Preseparation Period

Mothers and infants were monitored intensively during the 2 weeks before mothers were removed from the group. These preexperimental observations established baseline behavior levels and allowed generalizations concerning both infants' and mothers' activities and patterns of interaction within the group. Only data on the infants are reported here.

Levels of selected preseparation behaviors can be seen in figures 2–11, presented in two 1-week blocks, rather than on a daily basis, as they are for the separation period. Figures 12–31 indicate the degree of variability among individual infants in levels of performance of certain important social and solitary behaviors. These are discussed below.

Play, vocalization, and locomotion are especially important behaviors in evaluating social and emotional effects of mother loss on an infant (fig. 2). In general, 6- to 8-month-old infants play often and vocalize little. Figure 3 shows levels of solitary and social (including wrestle, chase, chase with contact, and synchronous) forms of play behavior. 6-month-old infants, both control and separation subjects, engaged in more social than solitary play throughout the observation weeks.

Approach behaviors by subject infants towards adult females during the preseparation period were particularly important. The 3 adult females *least* frequently approached by infants in the preseparation weeks were the 3 *most* often sought by infants to serve as substitute major caregivers during the mother's absence (fig. 4). In general, the mean frequency of approach to all other animals (fig. 5) was higher during the preseparation period than at any time thereafter.

The percentage of time that subjects were in contact with other animals during the preseparation period is shown in figure 6. Total contact time represents sums of: sit in contact, rest in contact, huddle, cradle, enclose, and passive support.

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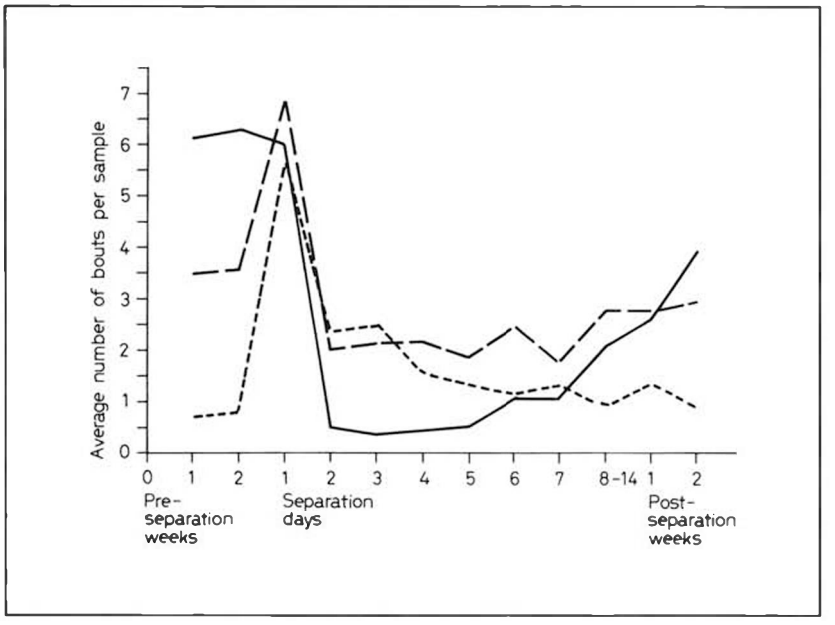


Fig. 2. Mean frequency per sample of play (——), vocalization (- - - -), and locomotion (— — —) for 9 completed separations.

Adults seldom protected infants of separation age. Figure 7 illustrates the frequency of protective behaviors including protective attack, protective threat, protective withdrawal, and protective interference relative to the infant. Threats (fig. 8) including glare threat, hand threat with or without contact, face threat, lunge threat, and chase directed towards and rejections of infants were also infrequent. They were, however, very noticeable as they were usually accompanied by loud infant protest. Figure 8 illustrates the mean frequency of different kinds of rejection directed at the separation subjects. Rejections, including contact deterrence, nipple deterrence, nipple withdrawal, punitive deterrence, avoidance, and play deterrence, were more frequent than protective or threat behaviors toward subject infants.

Infants were groomed approximately 18 times more often during the preseparation weeks than they groomed (fig. 9, 10). Their eating, mouthing, and manipulating activities are shown in figure 11; contrasts between preseparation and separation levels are discussed below.

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Separation Period

Although many effects of mother loss were observed on the first day of her absence, there was variability in the time of major impact on different behaviors. Separations began in late morning, but it was often several hours before distress or changes in infant behavior became evident. A few infants did not appear to miss their mothers until quite late in the afternoon – a time when infants usually sought to nurse and rest with their mothers. The amount of infant distress after the mother was taken from the group ranged from briefly expressed slight discomfort to great agitation and death. Results of the 2 separation weeks are presented in three parts: the general effects of the first 3 days of separation; individual behaviors, and adoptions.

The First 3 Days

There were dramatic changes in behavior at the start of separation. By late afternoon on day 1, all infants had begun searching for the lost mother, and vocalization and locomotion scores increased dramatically (fig. 2). Diagrammatically, play behavior is shown as remarkably unchanged for the day as a whole, but this seeming stability was due in part to the generally high frequency of play early in the day and in the first few hours after the mother was removed (fig. 3). Once the infant evidenced the onset of distress, its play became infrequent or stopped completely. There was great variability among infants as to the amount of distress shown and the consistency of its expression. Some infants began to search for the lost mother immediately and determinedly, while others did not appear to notice her absence for hours. Most infants showed intermittent distress – bouts of searching and calling would be interrupted or ended by play, for example, and would resume only later when something appeared to remind the infant of its loss. All infants sought care or attention from one or more adult females on the first day of mother loss. Not unexpectedly, some infants were much more effective than others in gaining this care. Infants that gave up quickly and did not persist in insinuating themselves into proximity or contact with the adult female of their choice were most likely not to gain a substitute caregiver. Clearly, certain females (especially BF, LT and ST) were preferred by infants and were approached far more often than other females (fig. 4). There was, however, no change in the mean frequency of approaches by separated infants to animals in general. Probably, in part at least, as a result of the infant's agitation and vocal searching, threats directed towards subject infants on day 1 of separation increased markedly in comparison with preseparation levels (fig. 8).

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The impact of mother loss is seen most clearly on day 2 of separation. The entire day was characterized, for all infants, as the day of greatest distress, with dramatic changes in the three major behaviors. This was a day of increased frequency of approaches to adult females and precipitous decline in play, vocalization, and locomotion from day 1 levels. Play, greatly decreased in comparison with preseparation levels, and in some instances ceased entirely on day 2. Vocalizations, less frequent than on separation day 1, were nevertheless higher than during the preseparation period and locomotion was much below preseparation levels. Although infant approaches to adult females were higher than during the preseparation weeks, there was a marked decrease in the mean frequency of approaches to other members of the group (fig. 4, 5).

Day 3 was marked by low levels of play, vocalization, and locomotion. The frequencies of play and locomotion continued depressed throughout separation and were slow to regain preseparation levels. Although vocalizations became less frequent after day 3, they remained higher than during the preseparation period.

Individual Behaviors

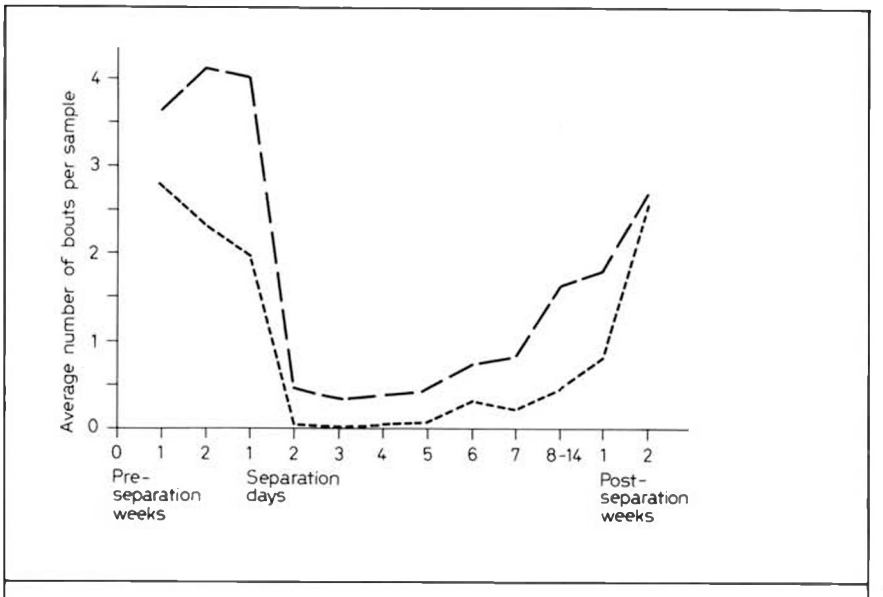
6- to 8-month-old infants did not vocalize frequently; preseparation levels indicated on figure 2 were the same as vocalization frequencies for control subjects. The greatest change in vocalization frequency occurred during the day 1 of separation, when the infant began to search and call for its lost mother. From an average of less than 1 call per 10-min observation sample, the level rose to more than 5 calls per sample on day 1. Although there was a decrease to slightly more than 2 calls per sample on day 2 of the mother's absence, this frequency was still significantly above preseparation levels. Frequency of calling gradually decreased from day 3 onwards and by the end of week 2 of separation was again only about 1 per 10-min sample. A one-way analysis of variance for repeated measures on vocalization revealed significant differences between days ($F = 14.06$, d.f. = 11.88, $p < 0.01$). Pairwise comparisons reveal significant differences between preseparation week 2 and separation day 2 (Newman-Keuls

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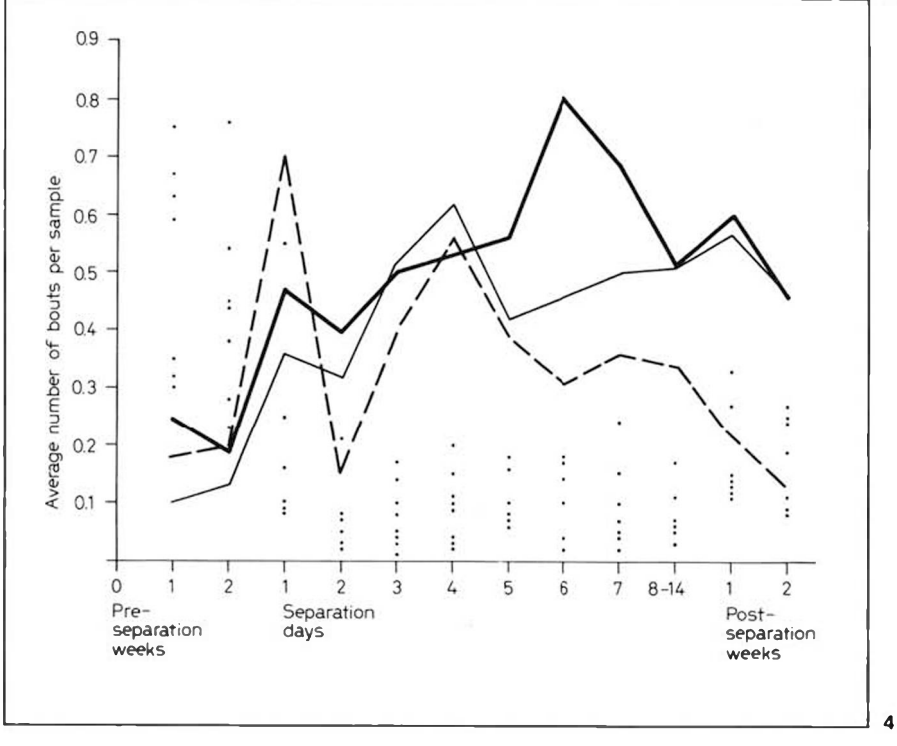
Fig. 3. Mean frequency per sample of solitary (-----), and social (– – –) forms of play for 9 completed separations.

Fig. 4. Frequency per sample of approaches to adult females by 9 separation infants. 3 adult females are shown: BF (▬▬), LT (——), and ST (-----). Dots indicate scores for other adult females in the groups.

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test, $p < 0.01$), between days 1 and 2 of separation ($p < 0.01$), but not between preseparation week 1 and postseparation week 2.

Locomotion was affected similarly (fig. 2). On day 1, when the mother was removed and the infant began to search for her, locomotion frequency increased from between 3 and 4 to about 7 times per sample. A one-way analysis of variance for repeated measures on locomotion demonstrated significant differences between days ($F = 18.74$, d.f. = 11.88, $p < 0.01$). There were significant differences between preseparation week 2 and separation day 2 (Newman-Keuls, $p < 0.05$), and between separation day 1 and separation day 2 ($p < 0.01$). Locomotion dropped sharply on day 2 to well below preseparation levels where it hovered throughout separation. Differences between preseparation week 1 and postseparation week 2 were not significant. Play, both social and solitary, was the most affected of all recorded behaviors. From a day 1 level that approximated preseparation levels, day 2 showed a precipitous drop in frequency of all kinds of play (fig. 2, 3). A one-way analysis of variance for repeated measures on play revealed significant differences between days ($F = 14.73$, d.f. = 11.88, $p < 0.01$). Differences in play frequency between preseparation week 2 and separation day 2 were significant (Newman-Keuls test, $p < 0.01$). Solitary play, always lower in frequency of occurrence than social play for infants of separation age, almost disappeared and did not reappear until the end of separation. Social play almost ceased on day 2 of the mother's absence. These forms were slow to regain preseparation levels and did so only during the postseparation period. Differences in play frequency between preseparation week 1 and postseparation week 2 were significant at $p < 0.05$ (Newman-Keuls). It should be noted that most social play recorded for separation days in figures 2 and 3 was initiated with the subject infants by other immatures in the group. The infant, bereft of its mother, seemed little inclined to play in the face of such a loss.

Figure 4 represents mean frequencies of approach to adult females by separation infants. The 3 adult females with highest scores were females most often chosen as substitute caregivers by infants during the 2 weeks of their mother's absence. These 3 females (BF, LT, and ST) were least often approached during the preseparation period, and most often approached during the separation weeks, according to the Spearman-Rank correlation coefficient computed for preseparation week 1 and the separation period ($r_s = -0.77$, $p < 0.01$). A similar trend was evident between preseparation week 2 and the separation period, although it did not reach statistical significance. Adult female approaches to infants (fig. 14, 18, 22, 26, 30) were low throughout the preseparation period and did not increase significantly during the separation period. Clearly, it was

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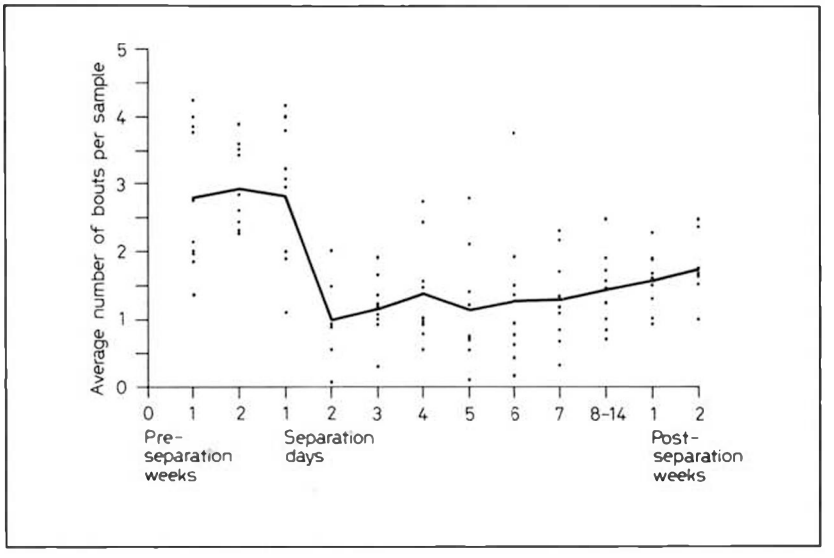


Fig. 5. Mean frequency per sample of approaches to all other animals by 9 separation infants. Dots indicate scores for individual infants.

the infant that approached and was responsible for maintaining contact with the female of its choice.

Figure 5 shows the mean frequencies of approaches by infant subjects to all other animals. A one-way analysis of variance for repeated measures of approaches revealed significant differences between days ($F = 14.0$, d.f. = 11.88, $p < 0.01$). Day 2 of separation was markedly by a significant drop in approaches in comparison with day 1 (Newman-Keuls, $p < 0.01$), and the level increased only slowly thereafter. Differences in approaches between preseparation week 2 and separation day 2, and between preseparation week 1 and postseparation week 2 are significant (Newman-Keuls, $p < 0.01$). Approach scores reflected the generally decreased infant interactiveness during separation.

Figure 6 illustrates mean contact duration, or percentage of total time that subject infants were in contact with other animals. Contact scores fell sharply on day 1 of separation. A one-way analysis of variance for repeated measures was significant at the 0.01 level. The Newman-Keuls test reveals that differences in contact between preseparation week 1 and postseparation day 2 were not significant. However, comparison of preseparation week 2 and separation day 1 was significant, as was the comparison of day 1 and day 2 of separation

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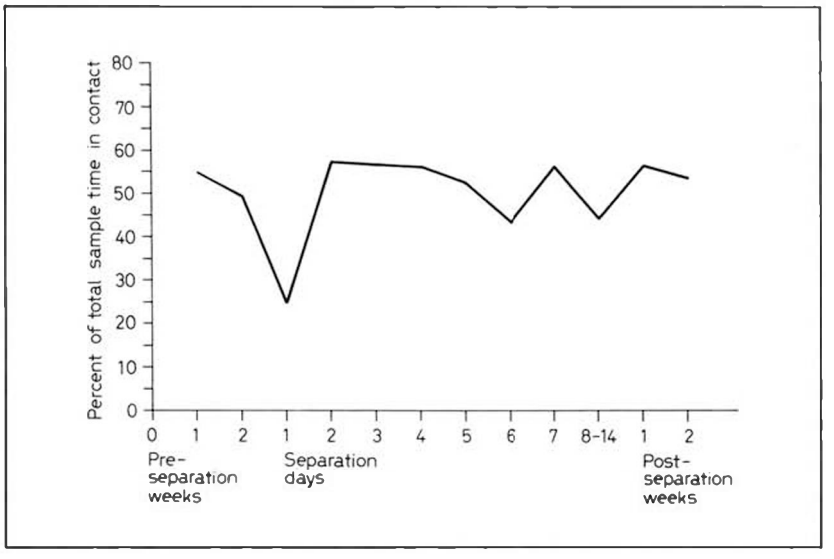


Fig. 6. Mean percent of total sample time that 9 separation infants were in contact with other animals in the group.

(Newman-Keuls, $p < 0.05$ and $p < 0.01$, respectively). Distress evident on day 1 of separation was reflected in low contact scores. Separation subjects did not gain substitutes for lost mothers until day 2 of separation at the earliest, when there was a marked increase in percentage of time in contact with other animals (fig. 6).

Separation age infants were in general no longer actively protected by their mothers. Figure 7 illustrates the low level of frequency of protection behaviors (including protective interference, avoidance, withdrawal, or attack). Although some infants were very agitated by their mothers' disappearance, other group members did not aid or protect these infants. Infants that succeeded in getting a substitute caregiver did not receive more protection from these females or any other animal than did infants that failed to gain a caregiver during their mothers' absence. The analysis of variance for repeated measures was not significant for differences in protective behaviors between any of the days shown on figure 7.

Threats from adults, including glare, hand, hand with contact, face, and lunge threats, chases, and bites, seldom were directed to infants of separation age. The day mothers were removed, infants received more threats per sample than did those whose mothers were in the group (fig. 8). In terms of total

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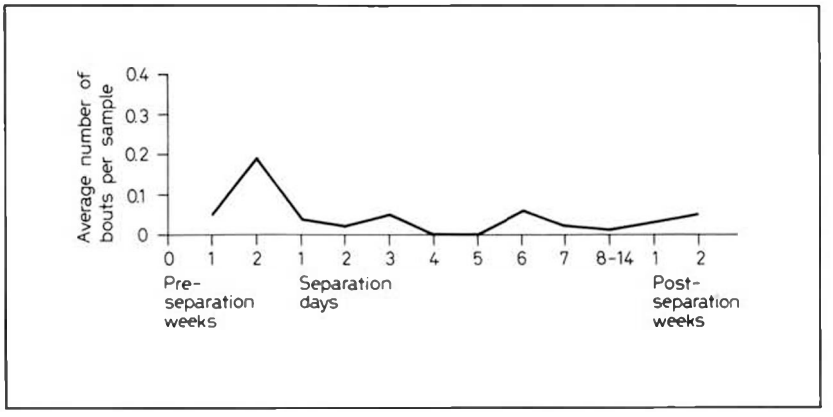


Fig. 7. Mean frequency per sample of protective behaviors directed towards 9 separation infants.

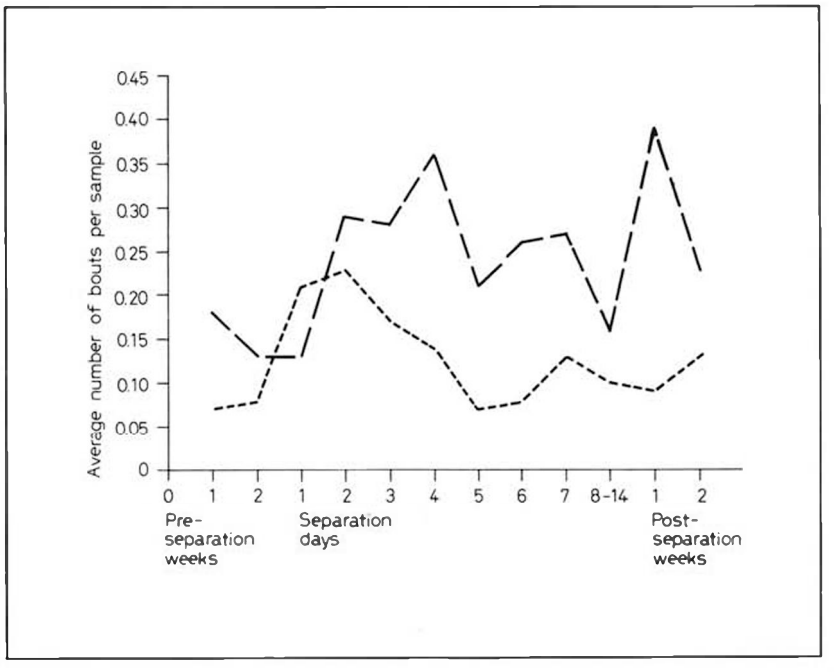


Fig. 8. Mean frequency per sample of threats (-----) directed toward and rejections (— — —) of 9 separation infants.

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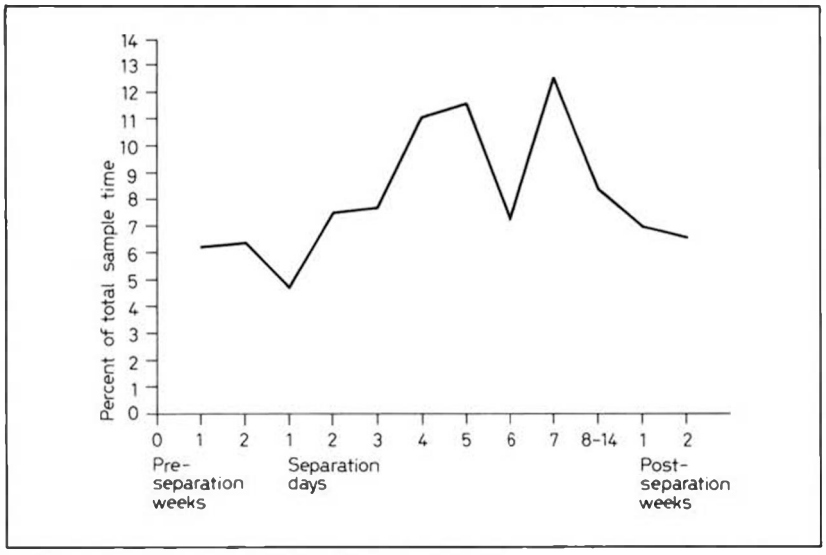


Fig. 9. Mean percent of total sample time 9 separation infants spent being groomed by other animals.

numbers, the increase did not result in many threats to separation infants since there were still less than 0.3 per 10 min sample. Fluctuations in threat frequency were relatively slight throughout separation and into the postseparation weeks. The increase in threats directed at the infant on day 1 of the mother's removal was due largely to the infant's annoyance of group adults in its persistent efforts to relocate the mother. Infants were rejected (as measured by contact deterrent, nipple deterrent, punitive deterrence, avoidance, and play deterrent) more often than they were threatened. Preseparation levels were higher than the low scores of separation day 1. Day 1 was low because many infants did not actively seek a replacement for their mothers. As day 2 arrived, most infants actively began to seek care and were, as a result, rejected more often. Rejection behaviors fluctuated throughout the separation and postseparation periods and were not affected in frequency by whether or not the infant gained a substitute for its mother. Pairwise comparison showed no significant differences in threat or rejection behavior between days on figure 8.

Although there was great variability, infants in general, groomed more during separation than in the preseparation weeks (fig. 9). Many grooming bouts began as adult females responded to an infant's attempts to maintain proximity

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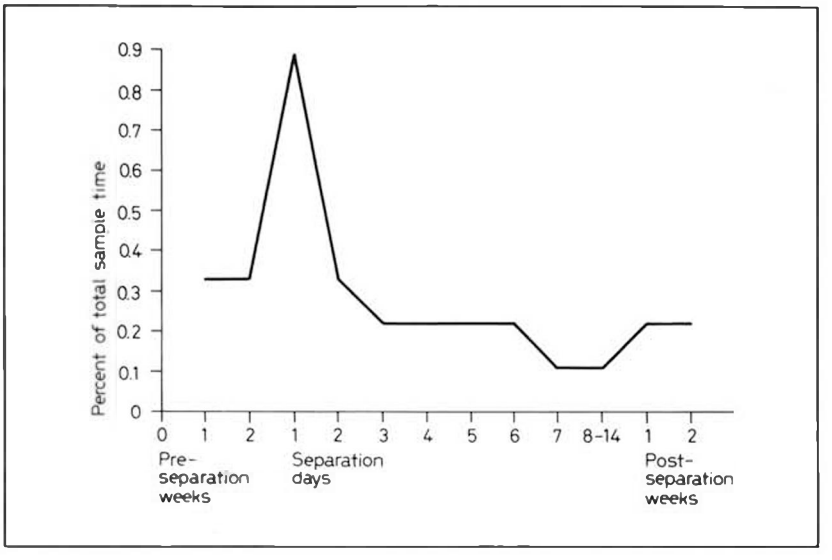


Fig. 10. Mean percent of total sample time 9 separation infants spent grooming other animals.

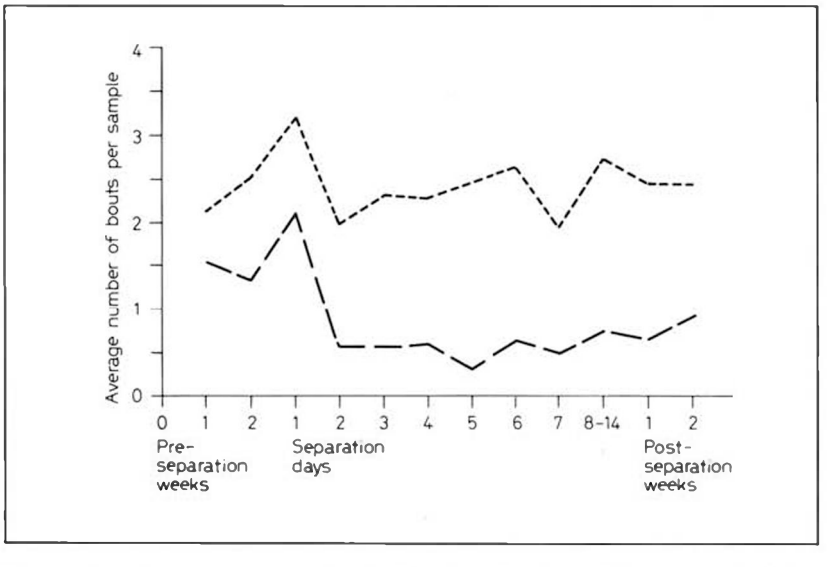


Fig. 11. Mean frequency per sample of eating (-----) and mouthing and manipulating (— — —) for 9 separation infants.

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or contact. Infants groomed others often on day 1 of separation, then suddenly groomed less during day 2, dropping to the preseparation level that day and going below it in the following days (fig. 10). The day 1 increase in grooming by infants reflected their attempts to stay near adult females in the late afternoon and early evening. Control infants did not groom often at this age, and their grooming attentions to adults were far less than adults' attentions to them; at most, infants groomed less than 1% of the time they were observed. Analysis revealed there were no significant trends in grooming behavior whether the subject was actor or receiver.

The amount of solid food infants ate during the preseparation, separation, and postseparation periods remained quite constant (fig. 11). As with the behaviors just noted, analysis of variance for repeated measures revealed no significant differences in frequency of eating between any of the test days. There was an increase on day 1 in the late afternoon after the mother was removed, but eating decreased on day 2 of her absence and fluctuated only minimally thereafter. Mother loss appears not to have had great impact on food intake for the surviving 10 subjects; all were similar in those scores. Mouthing and manipulating scores (fig. 11) also showed an increase on day 1 but then declined to below preseparation levels for the remainder of the experiment weeks. The difference between day 1 and day 2 of separation was significant (Newman-Keuls, $p < 0.05$), but all other comparisons between days were not.

Adoptions

All separated infants sought care from one or more adult females, but the frequency of approaches and persistence of effort varied. When adoptions occurred, they usually did so by day 3 of separation. The success of most infants in obtaining a caregiver probably was responsible for the change in levels of diagnostic behaviors such as the decrease in vocalization and locomotion on days 2 and 3 of separation (fig. 2). However, successful adoptions had little restorative effect on levels of play, which remained low during separation.

Of 9 infants that underwent completed separations, 7 adopted a female during separation (table I). Of the 7, only 2 went back to their mother when she was returned to the group. Of 9 completed separations, 4 infants (all males) returned to their mothers and 5 (3 males, 2 females) remained with the adopted caregiver. Of the 9 completed separations it would have been possible for 6 infants to have chosen to adopt adult females related to them, but none did.

The most appropriate description of the reaction of females who allowed infants to stay with them is that the former are passively accepting (fig. 32).

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Adults did not seek the infants. Some females selected by separation infants had their own infants; others did not. The infants of the substitute caregivers were of ages ranging from less than 1 month to 8 months. Persistence on the part of the subject infant paid off if the female of its choice had an infant of the same age or older than the subject. Conflict over access to the ventrum and to a nipple sometimes became acute, depending upon which infants were already associated with a particular adult female. At one period 1 adult female (BF) was maintaining as many as 3 infants simultaneously. In several instances when a separation subject adopted this female, the older of the 2 infants already with her left, leaving a younger infant and the subject to share her care-giving attention. BF allowed the male infant JA to stay with her when she already had her own neonate (fig. 33). The latter died, leaving JA sole claim to her ventrum. JA was joined by RA when the latter's mother was taken from the group, and JA and RA were joined $1\frac{1}{2}$ months later by CO when her mother was removed. After 1 week, JA left, but RA and CO remained with BF for months.

Although some of the adopting females probably were not lactating, every female selected by a separation infant allowed the infant to nurse. There appeared to be marked differences in the tolerance of some females toward neonates as compared to their tolerance of separation age infants. For example, one of the most frequently adopted adult females was harsh and punishing to her own neonates and to neonates of other females, and yet she was exceedingly passive with older infants. The care-giving qualities of colony females will be presented elsewhere.

The female infant SA, listed in table I as a control, left her mother when the mother began to reject her during weaning, and she too adopted the adult female BF. SA's shift from her mother to the other female was gradual and took longer than the more rapid attachments the separation infants formed. After some months with BF, the infant SA shifted to a third female because of competition with separation infant subjects for BF.

Postseparation Period

After 2 weeks' absence, the mother was returned to her social group. The most striking result of 9 completed separations was that 5 infants remained with the caregivers they had adopted during their mothers' absence (table I). Anticipated reunions, in over half the cases, did not occur. Most mothers did not actively seek out their infant nor make any obvious effort to solicit the infant's

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attention. If the infant returned, she accepted it and in no instance did she reject its approach or efforts to gain or maintain contact with her.

When a mother was returned to the group, she first reestablished relationships with other adult group members. This often involved active chasing and some hair pulling by the resident females. Wounding seldom occurred and was never serious enough to warrant removal of any animal for veterinary attention. The higher the status of the mother, the more strenuous the reentry skirmishes, although they were not necessarily longer in duration. Positions of power or rank were quickly reestablished, and by postseparation day 2, there were seldom any dominance interactions centered on the returned female.

Many infant behaviors returned to or near preseparation levels by the end of week 2 of separation, and the postseparation period was marked by gradual return to normal levels for other behavior patterns. Vocalization and locomotion approximated mean levels of frequency for control and preseparation subjects. Differences in locomotion and vocalization scores between preseparation week 1 and postseparation week 2 were not significant (fig. 2). For most separated infants, play remained below preseparation levels (significant difference between preseparation week 1 and postseparation level $p < 0.01$, Newman-Keuls test).

Infants generally did not approach other animals in the group as often as they did before separation, but scores for approach were within the range of those of control infants. The two adult females (BF and LT) approached most often by infants during separation continued to be most frequently approached by them during the postseparation period. The 3rd preferred female (ST) during separation was not approached as frequently in the postseparation period, and infants did not usually remain with her after their mothers returned.

Grooming (fig. 9), rejection and threats (fig. 8), and protection (fig. 7) directed to subject infants during postseparation weeks were unremarkable in frequency, and scores approximated preseparation levels. All infants continued to groom other animals only occasionally. Food intake, mouthing, and manipulating (fig. 11) continued at preseparation and separation levels.

Case Studies

Of the total of 12 infants separated, 5 represent the spectrum of responses to mother loss. OZ and PH both returned to their mothers after the latter rejoined the group, whereas CO and RA remained with the adult female caregivers they had adopted during their mothers' absence. DA died on day 3 of separation and represents the severest possible result of mother loss.

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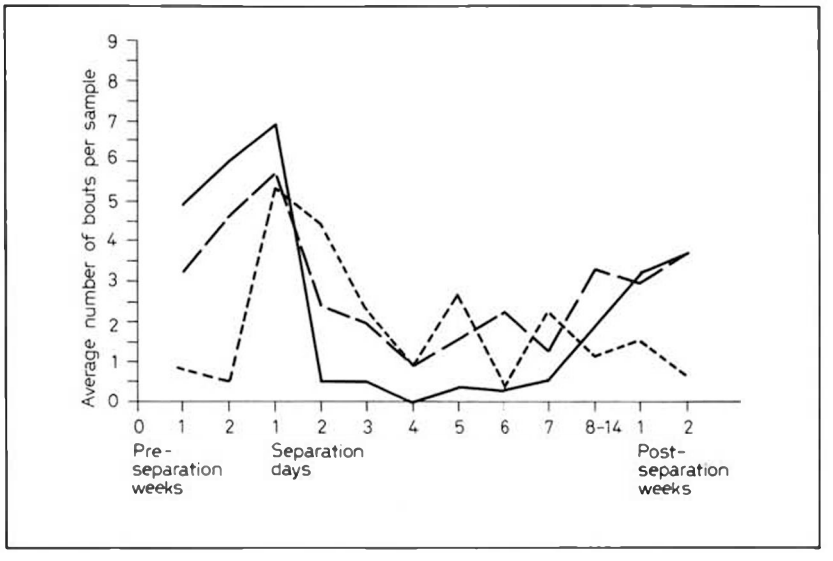


Fig. 12. Mean frequency per sample of bouts of play (——), vocalizations (- - - - -), and locomotion (— — —) for the male separation infant OZ.

OZ. OZ, son of an older multiparous female, HA, was 37 weeks old at the time of separation. His behavior profiles before separation (fig. 12) resembled those of the other subject infants (fig. 2). When his mother was removed, he sought care from several of the adult females in the group (fig. 13), but females approached him no more frequently than during preseparation weeks (fig. 14). On the day his mother was removed, OZ attempted to stay near the adult female LU more frequently than any other (fig. 13), and he continued to approach her until day 3, when he began to approach LT as frequently. On day 5, he approached LU again, and from day 6 to the end of separation, he approached her more often than any other female. When his mother was reintroduced, OZ immediately returned to her and approached her as often as he had during week 2 of preseparation.

OZ did not completely adopt a substitute caregiver. His contacts with LU and LT were infrequent in comparison to the scores of other infants during separation. The duration of time OZ spent in contact with the adult female LU was as great on day 7 of separation as it was with his mother after separation. The total amount of time OZ was in contact with other animals was greater during separation than before or after, when his contacts were largely limited to his mother (fig. 15). His contacts with animals other than adult females accounted for a portion of the scores illustrated on figure 15.

Figure 12 illustrates OZ's changes in play, vocalization, and locomotion during the 6-week experiment. On the day his mother was removed, his vocalization scores went up dramatically and significantly and remained high until day 3, when they went down towards preseparation levels. Vocalizations again peaked on day 5, the day he shifted his attention

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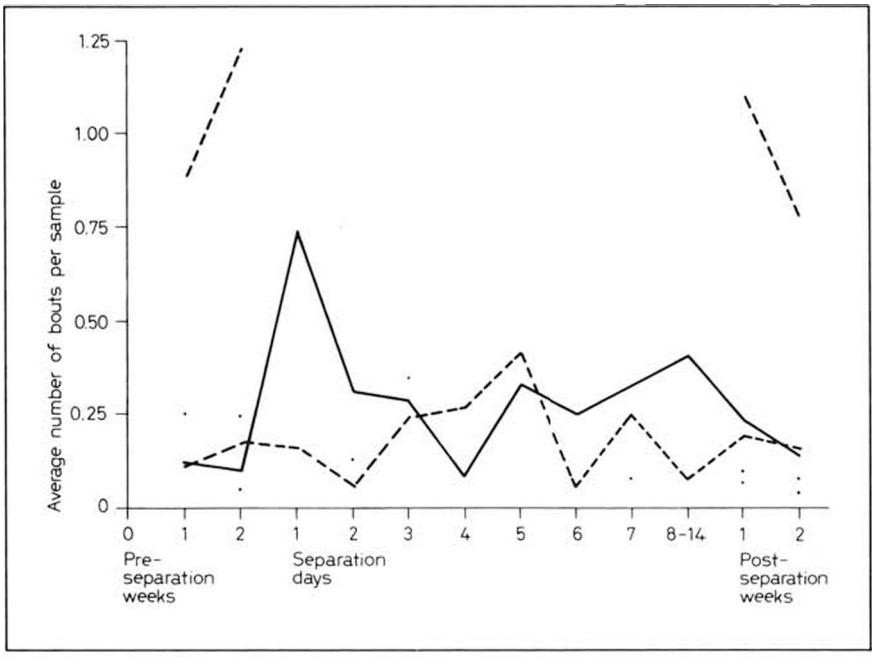


Fig. 13. Mean frequency per sample of approaches by the separation subject OZ to adult females LT (-----), LU (——), and his mother HA (– – –) upper left and right. Dots represent approaches to other animals at scores of 5% or more on 1 day or more.

from LT to LU. His vocalization frequencies fluctuated after that time, but by postseparation week 2 they were at preseparation levels.

Both play and locomotion dropped in frequency very sharply on separation day 2. Play remained depressed throughout the separation and increased only slightly during the following 2 weeks, reaching preseparation levels by the end of the postseparation period. Frequencies of locomotion fluctuated, but changes were not as pronounced as with play. Day 2 of separation saw a sharp decrease in locomotion which then fluctuated throughout the separation period and into the postseparation period. Locomotion was at or above preseparation week 1 levels during the last week of separation and in postseparation.

PH. PH, son of a young multiparous female, was 29 weeks old at separation. PH's preseparation week 2 was marked by a rather sharp decrease in the frequency of his play behaviors (fig. 16). Separation day 1 was marked by a sharp increase in play and locomotion and a lesser increase in vocalization. PH's mother was one of the more restrictive adult females in the colony, and when she was removed PH played much more often than she had allowed him to do during the preseparation period. However, day 2 of her absence was characterized by a strong decrease in both play and locomotion. PH had searched actively

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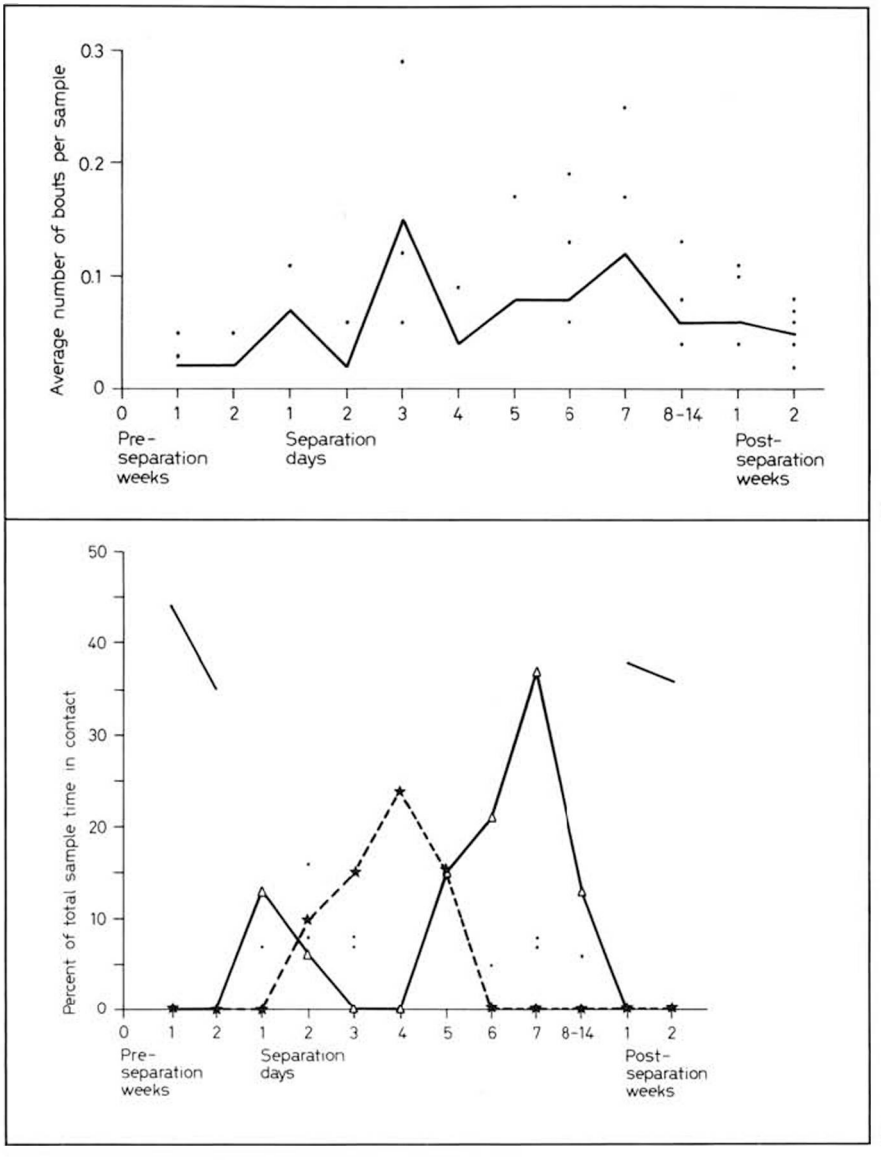


Fig. 14. Mean frequency per sample of approaches to the separation subject OZ by adult females in the group. Dots indicate scores for individual females.

Fig. 15. Percent of total sample time the separation subject OZ was in contact with the adult females LT, (*), LU (△), and his mother HA (——). Dots represent contact with other animals at scores of 5% or more for 1 day or more.

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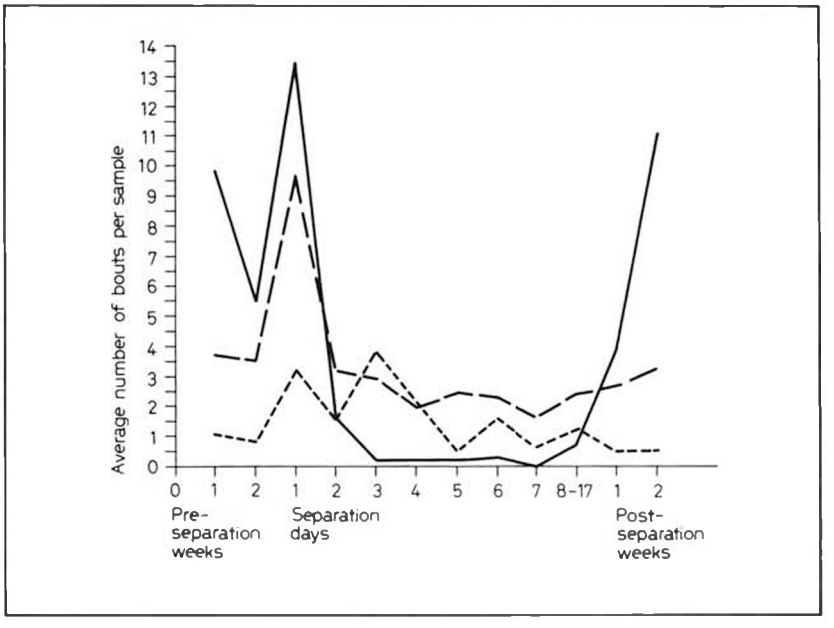


Fig. 16. Mean frequency per sample of play (——), vocalization (- - - - -), and locomotion (— — —) for the separation male infant PH.

for her the last part of the day 1, but on day 2 his locomotion scores were approximately the same as during the preseparation period.

Play almost dropped from PH's repertoire for the remaining days of separation but increased sharply when his mother returned. Locomotion, by day 3 of separation, was decreased in frequency in comparison with day 1 and hovered about preseparation levels thereafter.

PH was not able to gain a substitute for his mother, although he approached several adult females, especially ST, throughout the separation period (fig. 17). Like other subject infants during separation, PH sought and maintained contact with both male and female adults and immatures. He stayed in contact with ST more than with other females except on day 7 (fig. 19). Females did not approach PH significantly more frequently during separation than during the preseparation period (fig. 18). When PH's mother was returned to the group, he went directly to her – she did not approach him. His contact scores with her in the postseparation period were lower than they had been during the preseparation period.

PH ate more than did any other infant during separation, and he was considerably fatter at the end of separation than during the preseparation period. It is interesting to note, however, that the increase in food intake was not reflected in greater frequency of eating. It was not possible to make a simple translation of frequency of eating to the amount of food

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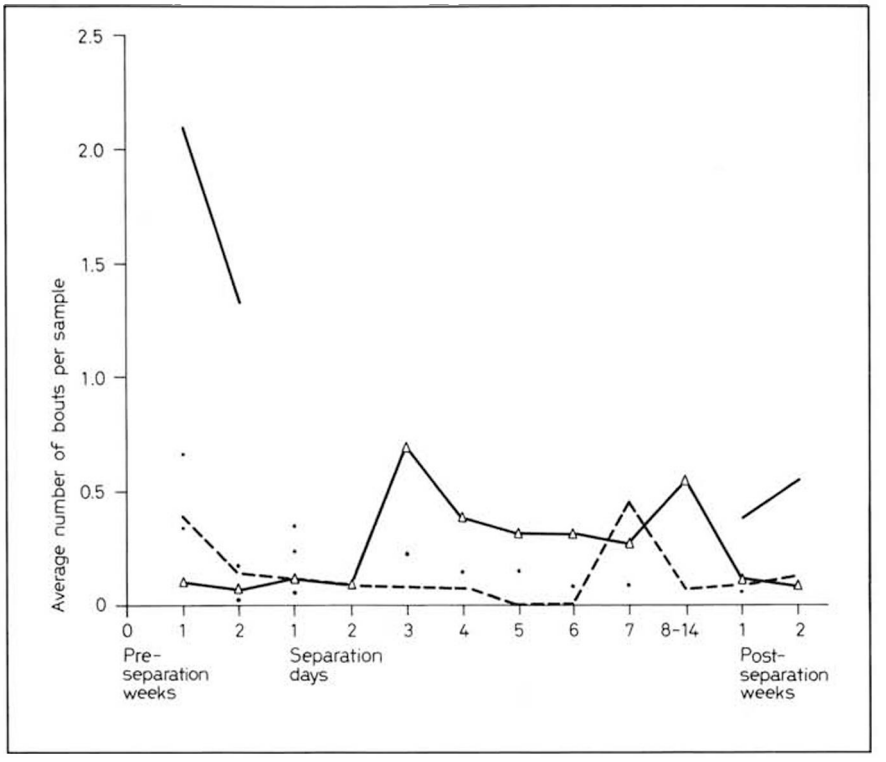


Fig. 17. Mean frequency per sample of approaches by the separation infant PH to adult females BF (- - - - -), ST (△), and to his mother LL (——). Dots represent approaches to other animals at scores of 5% or more on 1 day or more.

actually ingested. Activity levels also must be considered in weight gain or loss, and PH's final corpulence was doubtless due in part to his reduced activity level. Other infants with similarly reduced activity levels and normal eating scores did not gain weight.

CO. CO, daughter of an older multiparous female, was 34 weeks old at the time her mother was removed from the group. On day 1 of separation, although CO approached a number of adult females, she concentrated on ST and BF in an attempt to gain care-giving attention (fig. 21, 23). Although she went to ST more frequently than to BF on days 1 and 2, on day 3 of separation she approached BF more frequently than ST. After day 5 of separation CO stayed with BF for the rest of the separation and postseparation periods (fig. 23).

Figure 23 shows that CO had relatively high amounts of contact with the immature male RA from day 6 of separation. This was true because BF, the adult female selected by CO to serve as her major caregiver from day 6 of separation, had previously been chosen by

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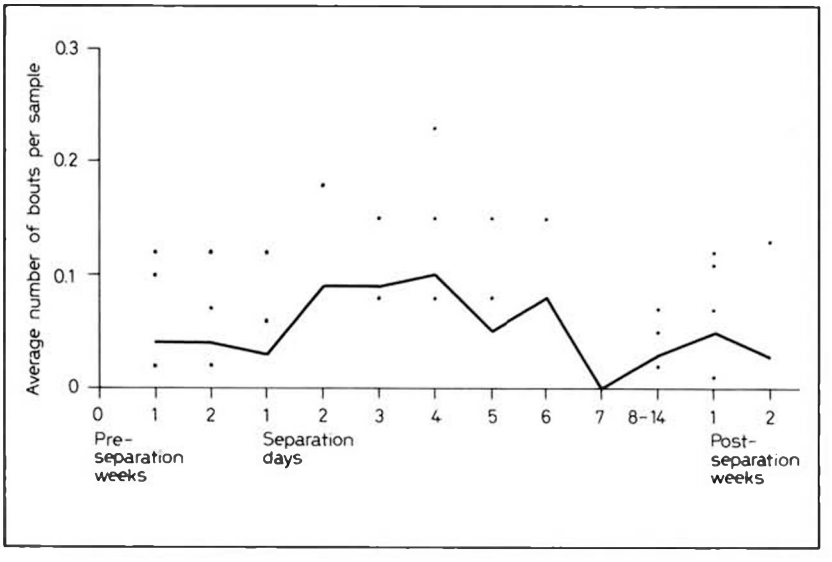


Fig. 18. Mean frequency per sample of approaches to the separation infant PH by adult females. Dots represent scores for individual females.

RA as his major caregiver when his mother was removed, and he had remained with BF for months after his postseparation period, as CO was also to do. The short line above weeks 1 and 2 of preseparation on figure 23 represents CO's approaches to her mother – the highest scores for CO's approaches to any female during that 2-week period. During postseparation when the mother (WA) was again in the group, CO did not approach her more frequently than she did any other adult female (fig. 23). CO's postseparation approaches to the female BF were almost at the level of her approaches to her mother during the preseparation period. No adult female in the group approached CO significantly more often during separation than during the preseparation period (fig. 22).

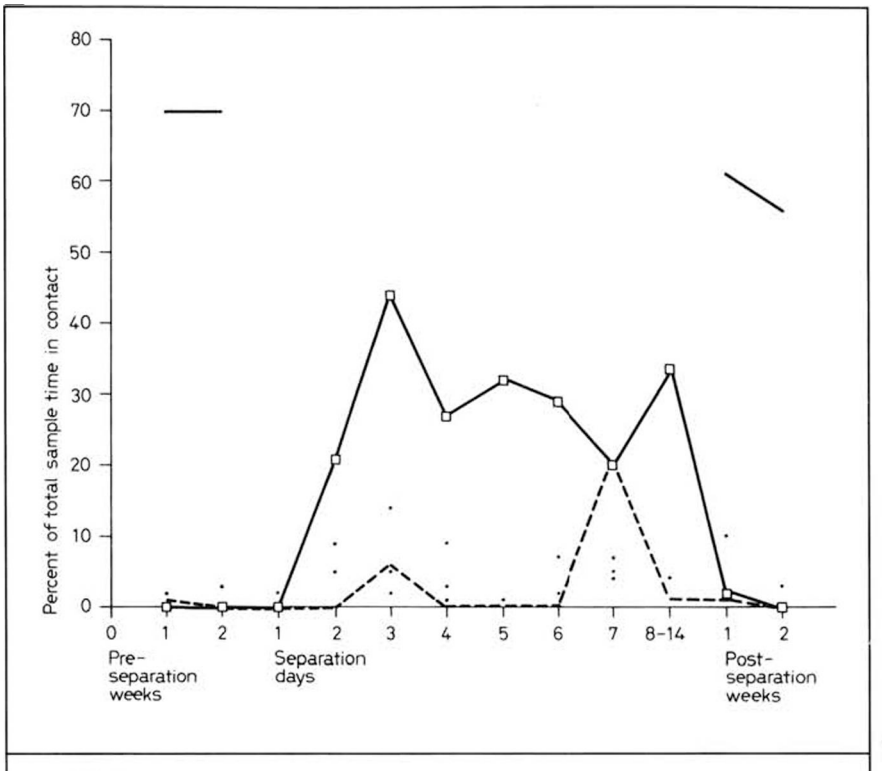
CO did not play for 2 days after her mother was removed (fig. 20). However, by the end of the postseparation period her scores had returned to preseparation levels. CO responded to the loss of her mother on day 1 with a sudden rise in vocalization and locomotion. She actively searched and called for her missing mother during late afternoon

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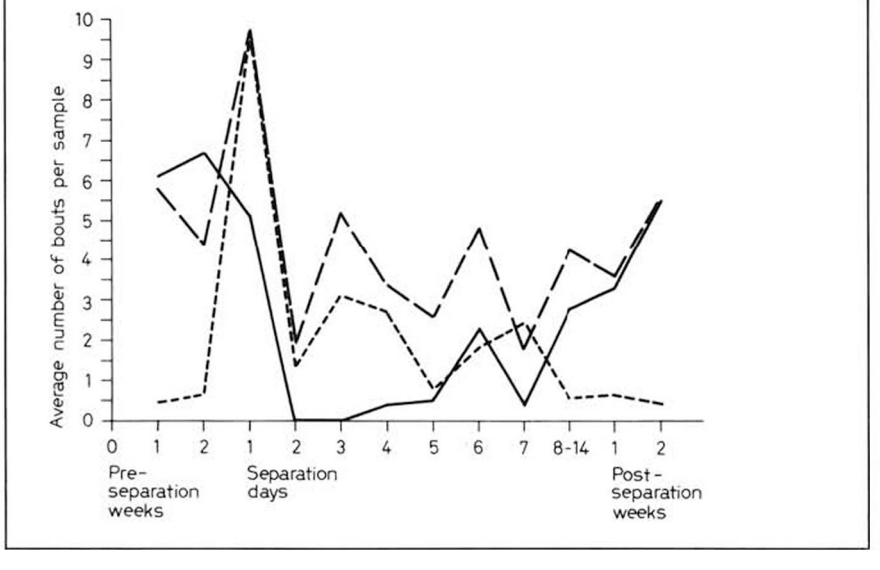
Fig. 19. Percent of total sample time the separation subject PH was in contact with other animals. Adult females BF (- - - - -), ST (□), and his mother LL (——) are shown. Dots represent other animals with scores of 5% or more on 1 day or more.

Fig. 20. Mean frequency per sample of play (——), vocalization (- - - - -), and locomotion (— — —) for the separation female infant CO.

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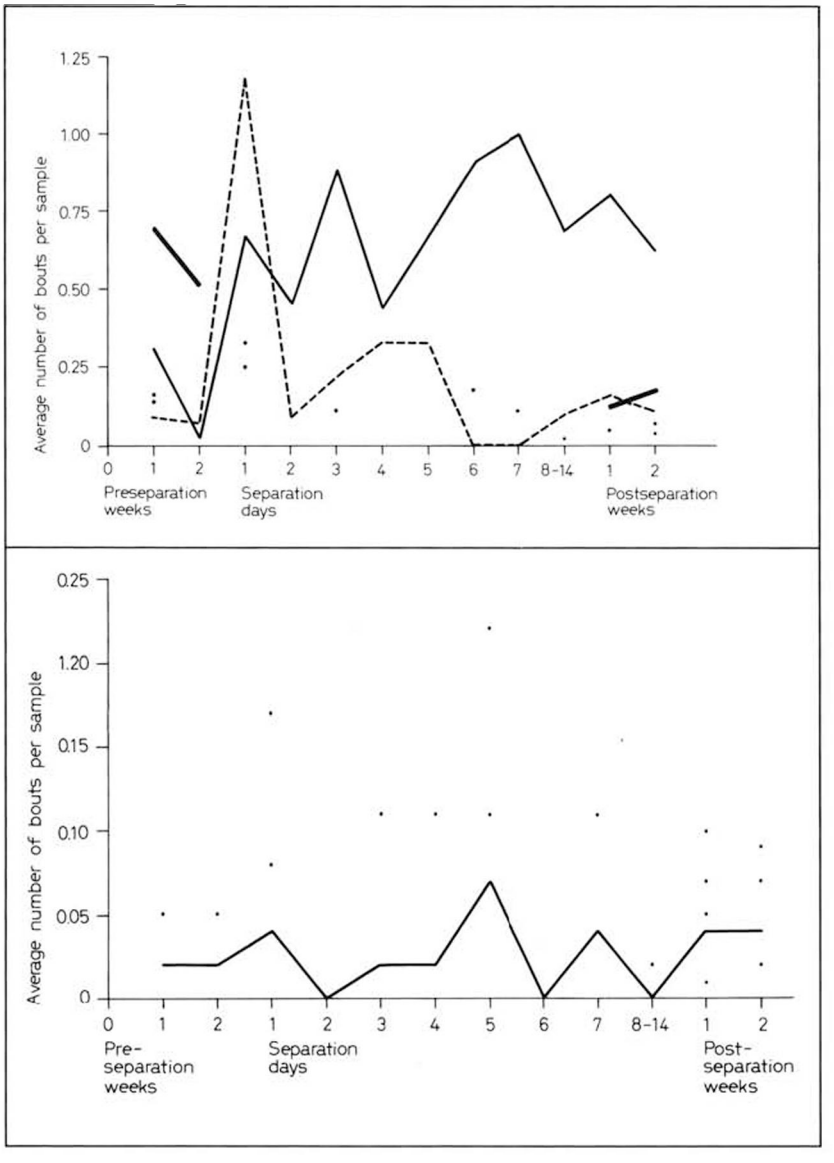


Fig. 21. Mean frequency per sample of approaches by separation subject CO to the adult females BF (——), ST (- - - -), and her mother WA (▬▬). Dots represent approaches to other animals at scores of 5% or more on 1 day or more.

Fig. 22. Mean frequency per sample of approaches to separation subject CO by adult females in the group. Dots indicate scores for individual females.

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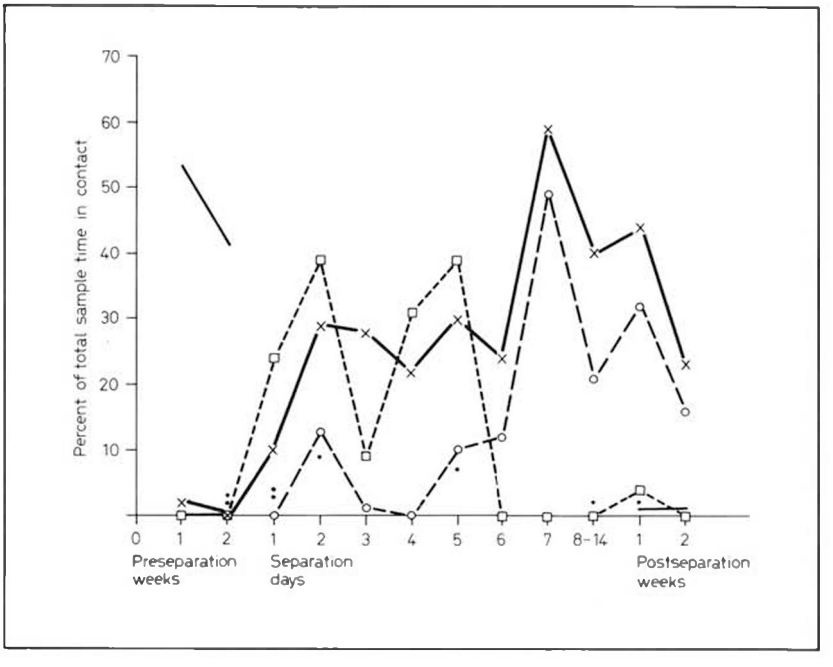


Fig. 23. Percent of total sample time the separation subject CO was in contact with BF (×), ST (□), and RA (○), and her mother WA (——). Other animals with scores of 5% or more on 1 day or more are shown as dots.

and evening. Her levels of vocalization fluctuated during week 1 of separation, but by week 2 they returned to and remained at preseparation levels. Locomotion scores reflected CO's agitation and searching on day 1 of separation but dropped on day 2, only to increase and decrease several times thereafter. Postseparation week 2 locomotion scores were the same as those of preseparation week 1 (fig. 20).

RA. The male infant RA's mother was removed from the group when he was 38 weeks old. RA was the second pregnancy but the first live birth of the young adult RP. Figure 24 shows a typical pattern of impact of mother loss at the beginning of a separation period with, for example, play dropping markedly and recovering slowly. Unlike those of the infant CO, RA's play scores did not return to preseparation levels by the end of the postseparation period. RA's locomotion scores, after the initial peak on day 1 and drop on day 2, rose and fell less dramatically during the remainder of separation, but by postseparation week 2 they were at the level of preseparation week 2. RA's vocalizations rose in frequency on the day his mother was removed; they returned to preseparation levels on day 4 but rose again later in the separation and went higher after his mother's return. Postseparation week 2 vocalizations were within the range of preseparation levels for this infant.

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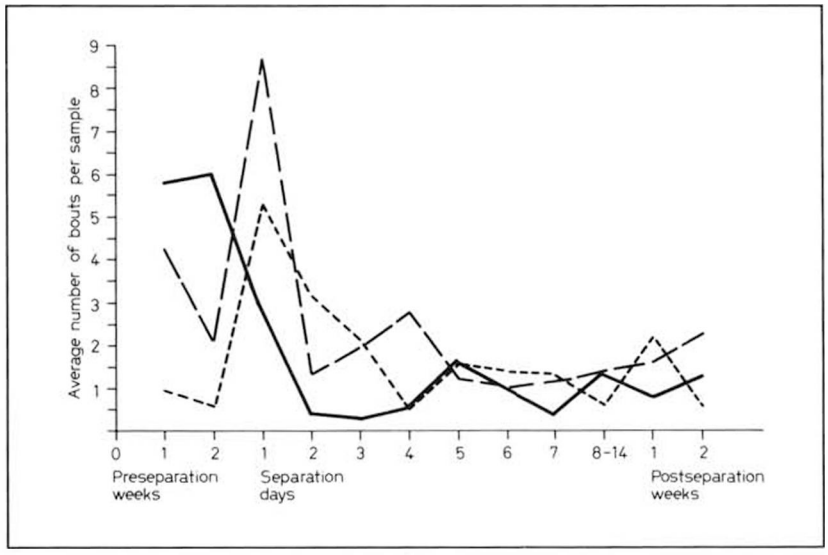


Fig. 24. Mean frequency per sample of play (——), vocalization (- - - -), and locomotion (— — —) for the separation male infant RA.

Figure 25 shows that on day 1 of separation, although RA approached a number of adult females, he concentrated on BF. He maintained high frequencies of approach to her through the separation and postseparation periods. His contact scores also reflect his strong preference for BF (fig. 27). In fact, RA maintained longer contact with BF than he had with his own mother, and his total contacts with all animals were higher throughout the separation and postseparation periods than during the preseparation period when he was with his mother. Neither BF nor the other females in the group approached RA often (fig. 26). It should be noted that RA did not often approach his mother after she was returned to the group (fig. 25). Instead, he remained with his adopted caregiver, BF, even though she had a neonate of her own.

DA. The male infant DA was 30 weeks old when his mother, a multiparous adult female, was removed from the group. His preseparation scores for vocalization, locomotion, and play (fig. 28) were within the normal range for infants this age, and the sharp increase in all three scores on day 1, when his mother was removed from the group, is a typical reaction to the separation. Play, vocalization, and locomotion frequencies decreased to zero on day 2. At the beginning of day 2 he was extremely subdued and by the end of the day he appeared to be very tired, ignoring the frequent play invitations by other infants in the group. Early on day 3 it was obvious that he was in trouble because he did not eat and moved very little. His lethargy was punctuated by an occasional soft vocalization, and he was dead by early afternoon.

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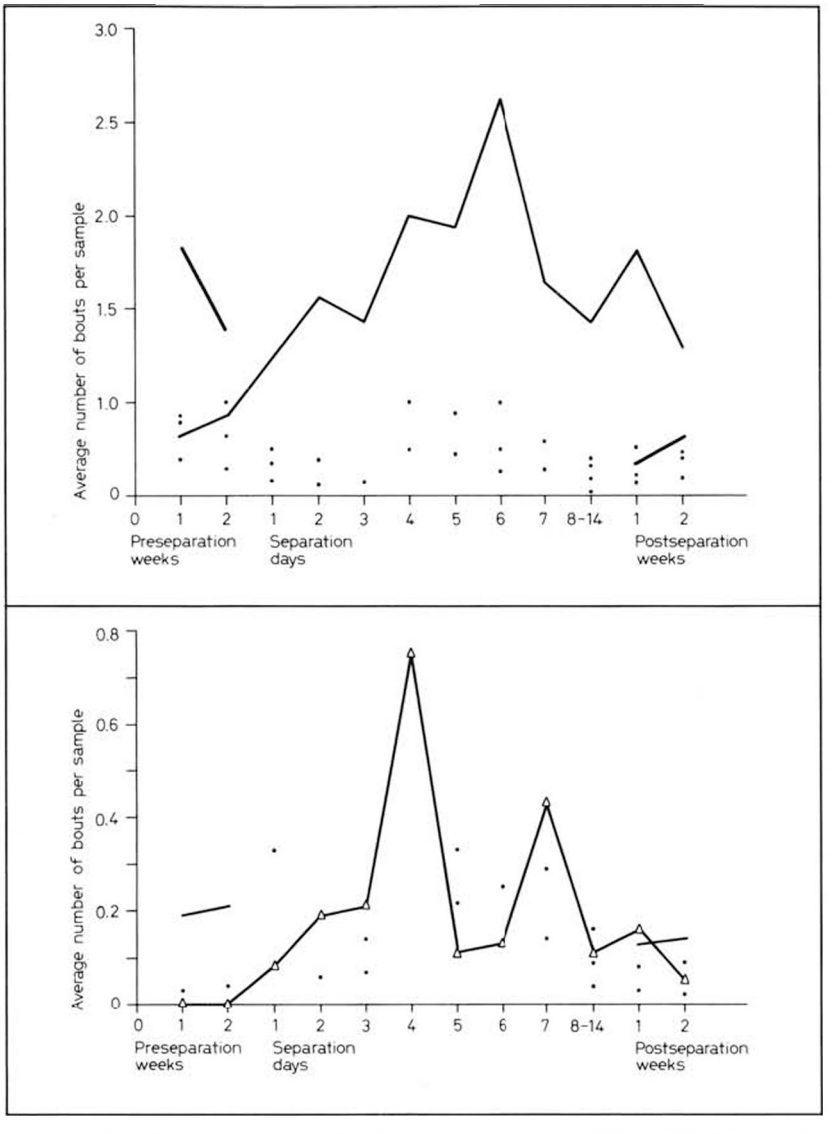


Fig. 25. Mean frequency per sample of approaches by separation subject RA to the adult female BF (——) and to his mother RP (▪▪▪▪▪). Dots represent other animals with scores of 5% or more on 1 day or more.

Fig. 26. Mean frequency per sample of approaches to the separation subject RA by the adult female ST (△) and his mother RP (——). Dots represent other animals with scores of 5% or more on 1 or more days.

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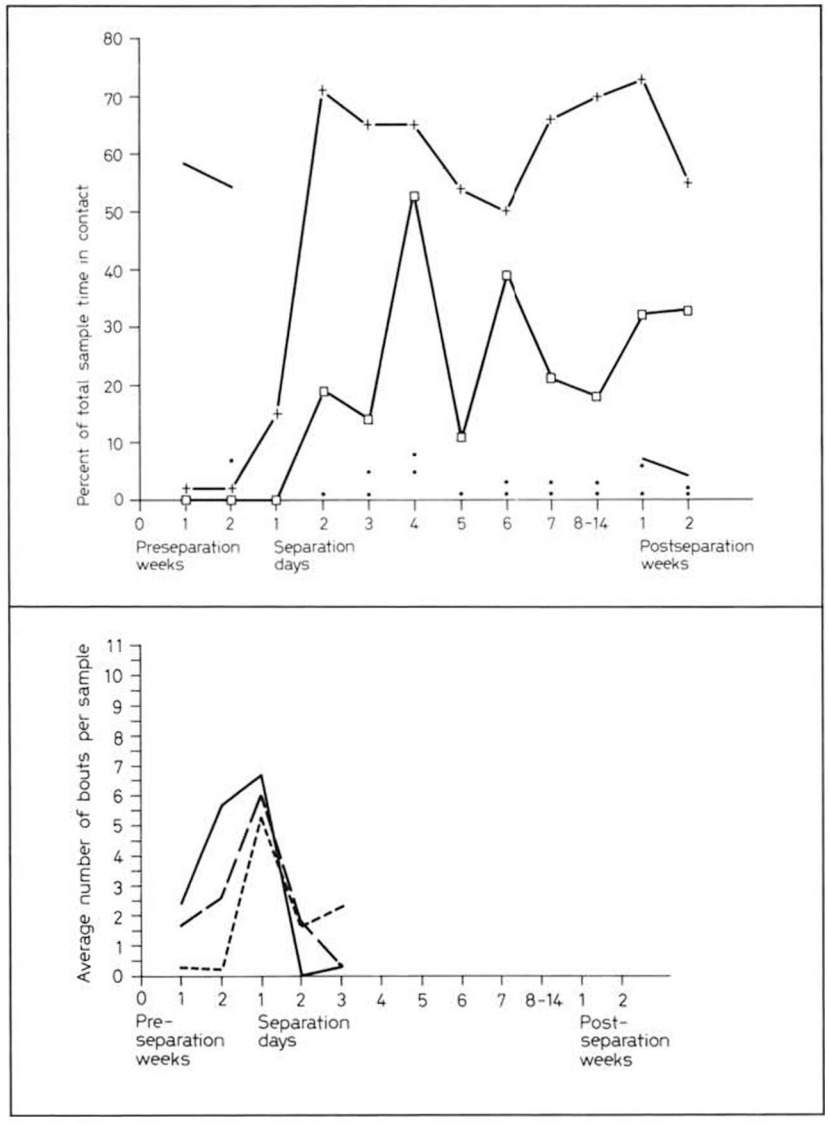


Fig. 27. Percent of total sample time the separation subject RA was in contact with the adult females BF (+), his mother RP (——), and to the male infant JA (□) who was associated with the female BF. Dots represent other animals with scores of 5% or more on 1 or more days.

Fig. 28. Mean frequency per sample of bouts of play (——), vocalization (- - - - -), and locomotion (— — —) for the separation male infant DA which died on day 3 of separation.

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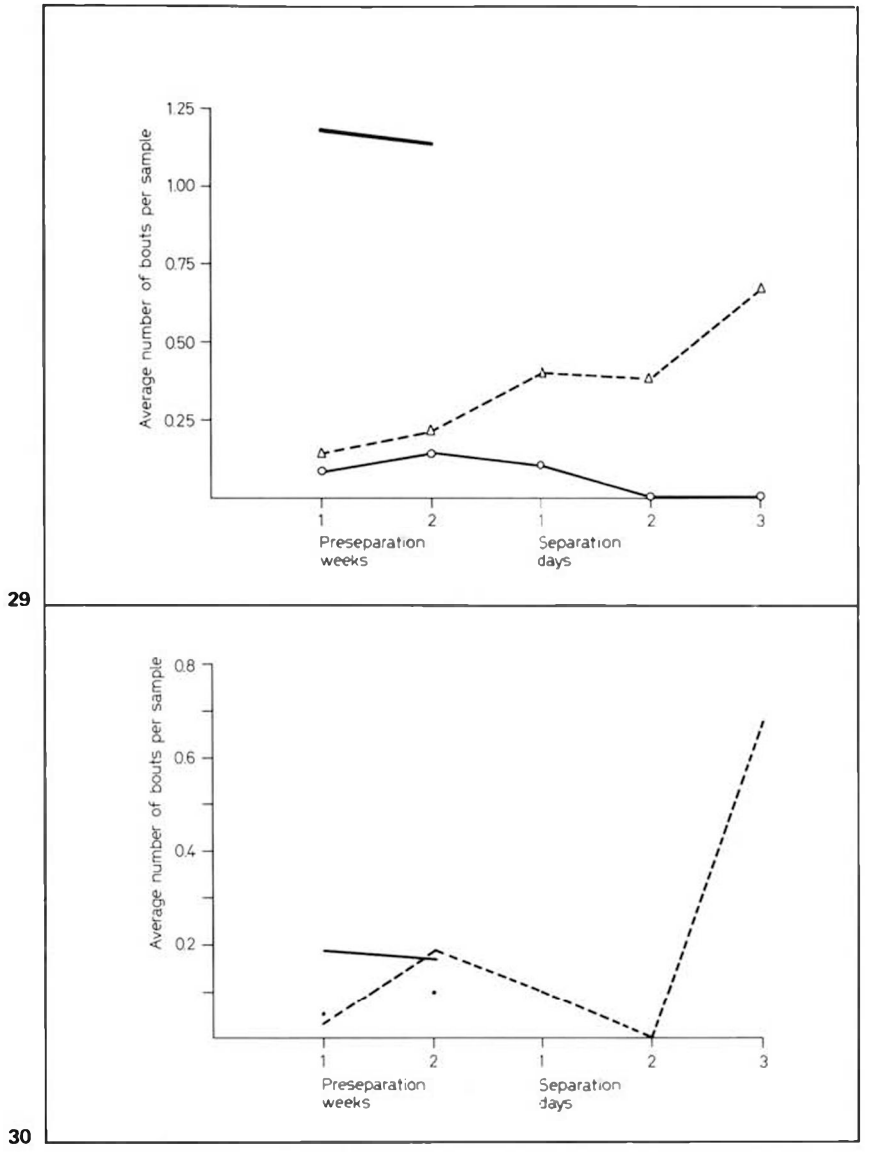


Fig. 29. Mean frequency per sample of approaches by separation infant DA to the adult females GI (○), MO (△), and his mother LU (——).

Fig. 30. Mean frequency per sample of approaches to separation subject DA by the adult female MO (-----) and his mother LU (——). Dots represent other animals with scores of 5% or more on 1 day or more.

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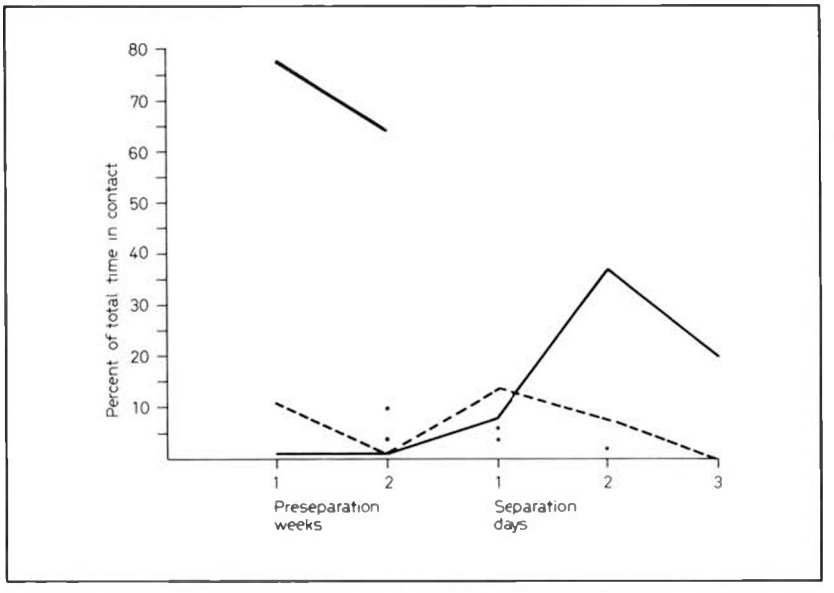


Fig. 31. Percent of total sample time the separation subject DA was in contact with the adult female MO (——), his mother LU (▬▬) and with the juvenile male OZ (- - - - -). Dots represent animals with scores of 5% or more on 1 day or more.

DA did seek and gain contact with MO on the 3 days of separation he survived (fig. 29, 31). MO approached DA more often on day 3 than at any other time (fig. 30), but even then his contact scores with her were low (fig. 31) compared with his preseparation scores with his mother (fig. 31). DA's inability to obtain the care he sought undoubtedly contributed to his great distress and death.

Discussion

This research was designed to identify and investigate significant factors in the relationship between a langur infant and the most important member of its social environment. Major aims have been to provide new and supplemental insights into the spectrum of potential responses to mother loss in monkeys, and to provide specific data illustrating how the Colobine multiple caregiver pattern influences the effects of mother loss and potentially increases the likelihood of an infant's survival when its major caregiver dies.

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Major Results

Four major results may be outlined.

(1) All 12 infants showed changes in their behavior during separation. Sensitive behaviors included play, vocalization, locomotion, approach and contact behaviors. Changes in behavior frequencies ranged from slight to pronounced. Reactions ranged from minimal to very severe, and 2 infants died within 3 days of the mother's removal from their social group.

(2) All infants sought contact and care from other adult females during separation, and most showed strong preferences for certain females. Importantly, most infants adopted a female to serve as the major and usually only caregiver during the mother's absence. On occasion, infants did seek care from peers.

(3) Adoptions of substitute major caregivers were infant-initiated. Some adult females, even though they had infants of their own, permitted adoptions but did not initiate them. Other adult females rejected infant advances and thus were not potential caregivers.

(4) Of 9 completed separations, 5 infants elected to remain with their adopted caregivers after their mothers were reunited with the group.

The macaque model of infant depression can be used as a heuristic aid in measuring and comparing factors important to understanding langur response to mother loss. For example, it has been demonstrated that developmental sequences of the depression syndrome reported for different species of macaques are exceedingly similar. There is agreement from both field and laboratory work concerning the timing and sequence of stages [Hansen, 1966; Harlow et al., 1963; Hinde and Spencer-Booth, 1967; Imanishi, 1963; Kaufman, 1977; Simonds, 1965].

Against this background of general macaque similarity, there are important and striking differences between bonnets and pigtails, expressed most dramatically in the reactions of infants to separation from the mother. Kaufman and his colleagues removed mothers from infants of both species when the infants were between 4 and 6 months of age.

Macaque reaction to separation was observed to consist of three phases: agitation, depression and recovery [Kaufman, 1967b and personal commun.]. *Agitation:* the infant hyperactively searched for its lost mother, moving more from place to place, and vocalizing. Observations suggested that it was experiencing a state of agitated distress with feelings of apprehension and anxiety. The increase in vocalization and locomotion was true for both bonnet and pigtail infants; and in both, during day 1 after separation, infants were the objects of

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increased punishment by group adults. Pigtail infants showed the greater distress. They sought caregiving attention from group females, but instead of receiving it they became targets of some hostility. Bonnet infants showed agitated disturbance and also searched for attention from adult females in the group. In contrast to the pigtails' frustrated and thwarted seeking of attention, bonnets received care or comforting from group females. *Depression:* after the infants' agitation subsided, they evidenced some degree of depression. Movement and play decreased, oral behavior increased, and the animals often assumed a slumped posture. Pigtail infants demonstrated greater changes in behavior and deep depression, similar to the anaclitic depression Spitz [1946] has described for human infants who suffer loss of mother. The bonnets' depressive reaction was much milder, in part because bonnet infants were often adopted by adult females. The latter were accepting even when they had infants of their own. *Recovery:* by day 10 after the mother was removed, both pigtail and bonnet infants were moving around, although playing and oral behaviors were still depressed by preseparation standards. Pigtails were slower to recover than bonnets, but with only two exceptions they did. The return of pigtail mothers was followed by a great reunion, and for approximately 3 months mother and infant maintained a close relationship. When the bonnet mother returned, the reunion was much less dramatic, and some infants elected to remain with their adopted mothers.

Whereas pigtail macaques interact primarily within a maternal clan (a mother and her descendents), bonnets interact with all group members regardless of kin ties. Kaufman [1973] presented strong evidence that the differences in behavior of these closely related species, although based in the genotype, are stimulated by differences in ontogenetic experience, especially maternal behavior with the infant during development.

Kaufman and his coworkers planned experiments with mixed species groups containing both bonnet and pigtail macaques in order to increase or decrease the likelihood that an infant would obtain relief from substitute caregivers. When an infant reached separation age, all females of its own species are removed so that only females of the other species remained. One 5-month-old bonnet infant, after removal of his mother and all bonnet females from the group 'showed as profound a depressive response as we have seen. In fact only 3 pigtails showed a comparable amount of body collapse, and 2 of them subsequently died during separation' [Kaufman and Stynes, 1977, 1978].

Variations of depression, in degree and expression have been observed for different macaque species and in different contexts of separation [Kaufman

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1973, 1977; Kaufman and Rosenblum 1967a, b, 1969; Kaufman and Stynes 1978]. Depressive reactions have been produced by means other than loss of mother [Harlow et al., 1970; Suomi and Harlow, 1975]. If the peers of an infant are its only social companions and they are removed, the remaining infant is bereft and depressed [Bowden and McKinney, 1972; Erwin et al., 1971; McKinney et al., 1971]. More recent reviews were concerned with the effects of different combinations of separation and social deprivation [Hinde and Davies, 1972; Hinde and McGinnis, 1977; Lewis et al., 1976; Suomi et al., 1974; Sackett et al., 1976; Ruppenthal et al., 1976].

At best, observations of behavior provide only an approximation of the effects of loss on an infant. In addition, it will be necessary to monitor physiological changes that are associated with the infant's perceptions and reactions to loss. For example, the importance of heart rate and body temperature recordings is evident from those made on 10 unrestrained pigtail macaques during the early agitation reactions of mother loss and for several days after the infants were reunited with their mothers. 'The behavioral agitation reaction immediately following separation was accompanied by increases in both HR and BT. Beginning with the first night of separation, both HR and BT showed marked decreases from the base line. Whereas group mean HR and BT changes were maximal early in separation, behavioral indices of depression tended to be maximal later in the separation period. Reunion with the mother tended to normalize HR and BT in most infants' [Reite et al., 1978; p. 91].

Recent observations of a single pigtail infant adopted by a pigtail female immediately after separation (an unusual situation for pigtail infants) demonstrated that similar physiological changes still occurred after separation from the biological mother [Reite et al., 1978a, p. 370]. Physiological monitoring of surrogate-raised pigtail infants after removal of the surrogates suggests that 'the attachment bond to a cloth surrogate may well be different that the bond to a living mother in a social group' [Reite et al., 1978b, p. 427].

Langur Monkey Reactions to Mother Loss

Although it may take a few hours before a langur infant's behavior reflects a reaction to its mother's disappearance, the latter portions of day 1 of mother removal is marked by mild to acute infant distress. The first night of separation appears to be critical, because it is immediately after that time that the langur infant demonstrates the greatest changes in behavior. Our observations indicate

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that the first night of separation – the first in the life of the infant that it has not been with its mother – is extremely important in the development of the infant's feeling of loss and agitation. Day 2 was the most stressed day for all subject infants; it was the day on which infants displayed the most agitation. The severity of day 2 reactions did not appear to depend on how many hours the mother had been gone on day 1. On day 2 play almost ceased, and both locomotion and vocalization scores dropped precipitously. The infant no longer constantly searched for its lost mother, although it did vocalize more than before separation. Only in the most severely distressed infants that died after the mother was removed, did we observe an obvious disengagement from most interactions.

Although some details of langur response to loss differ from macaque reactions, the three stages of macaque reaction to mother loss do provide a framework against which the langur reaction may be compared. Agitated seeking typifies the langur's immediate response to loss. This if often but not always followed by some degree of social withdrawal, during which the young langur monkey appears to conserve its resources. In a few langur infants this stage of reaction is suggestive of depression. The response may well depend upon the social, emotional, and psychological development of the infant and its ability to manage stress and missing – and on the infant's ability to gain care from the other group females. Most langur infants, although affected by the loss, maintained fairly normal levels of activity and continued to pursue, as far as possible, their daily living patterns. Finally, all infants recovered except for the 2 that died soon after separation. For some infants this recovery was a slower process than for others. Unlike macaques, langurs do not respond to mother loss with increased orality or with a significantly greater amount of object manipulation.

It should be noted that infants leaving their mothers because of weaning do not demonstrate the changes in diagnostic behaviors characteristic of separated subjects. Nor do the weaned infants appear to experience the stress and 'dejection' that characterize some separated subjects.

Factors Influencing Infant Response to Separation

Of the many factors involved in langur response to separation, the data presented here suggest that the following four are especially relevant.

(1) Qualities of maternal care. The more a mother restricts her infant's social interactions during the preseparation period, the more distress the infant

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shows. These influences on the infant are often very subtle and their impact on the infant's attitude is gradual. The mother's fears of specific animals in the group seem to be mirrored in the infant's behavior when she is gone. An example of this was MO, whose female infant AN died on day 2 of separation. MO was exceedingly wary of the adult male in the group and quickly retrieved AN whenever he approached either of them. When MO left the group, AN showed great alarm and avoidance when the male was nearby. This fear increased her early agitation and intensified her withdrawal on day 2.

In general, mothers that are not restrictive, including several that actively seek to leave their infants, offer their infants correspondingly greater opportunities to interact with group members. Our observations reveal that mothers do not necessarily treat successive infants in the same manner. One adult female was so restrictive of and punitive towards one of her infants that it died. She treated her next infant gently, and it was one of the least distressed of all separation infant subjects.

(2) Availability of accepting adult females. If there are such females in the group, subject infants quickly approach and try to remain with one of them. Quality of care seems to be almost irrelevant to the strength of effort a subject infant makes to remain with any of the caregivers. Of the 3 preferred females, none had high interactive scores with the subject infants during the preseparation periods of observation (fig. 4).

Some adult females, whether lactating or not, never allowed a separation subject to remain near or with them and were not available as a source of care for any infants other than their own.

(3) Adopted adult females' reproductive state. Subject infants showed strong preferences for females that were lactating. One of the preferred females, ST, was not lactating and was abandoned by subject infants when they had an opportunity to switch to BF, a lactating female.

(4) Presence of other infants. The number of same- or near-aged infants in the group is important for two reasons. First, they present competition for available adult female caregivers. Second, when there is a peer play group available, the subject is more frequently solicited to engage in social play or to interact than it would be if there were few or no potential play partners. Other infants continue to solicit play from the subjects at the same rate during separation as before, regardless of whether or not the subject infant responds. Play solicitations can be very tiring to the experimental subject, depending of course upon how the subject reacts – wheter it ignores invitations or actively retreats.

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(5) Perceptions by the subject infant. Major factors in the amount of distress after mother loss appear to lie within the infant itself and are subject to speculation only at this point in the research. All subjects had adult females nearby to remind them of their mother and her absence. Only certain infants reacted strongly to the absence of the mother, and the range of reactions was, as discussed, from mild to fatal. The extent of distress and its form appear to relate to the resources of the infant as well as to factors in the social environment, such as the availability of accepting females.

Adoptions

We anticipated that if a langur infant were able to gain care from an adult female in the group while its mother was gone, it would be markedly less distressed. As expected, all infants sought care and most – 7 out of 9 – were able to stay with one female for all or part of the mother's absence. However, the anticipated relief of symptoms of stress did not occur as a result of acquiring a substitute care-giving adult.

Infants also sought contact with other group members and they obtained it – in some instances more contact than with the mother while she was still in the group. In general, persistence in seeking to be near or touching others was rewarded. Figure 6 shows that the mean percentage of time separated infants were in contact with other animals dropped significantly only on day 1 of separation. There were fluctuations from day to day in the amount of time the subject infants gained contact with other animals, but the variations were not significant. Although the total time remained relatively the same, the subject infants had contact with more animals during the separation weeks, thus affecting the total contact scores for each day or week. Close inspection of figures 15, 19, 23, 27, and 31 reveals that of the 9 females of the group (listed in table I), some seldom if ever appear in the contact scores for subject infants.

The amount of contact with the mother before separation does not permit an assured prediction of the outcome of separation. (Details of approach, proximity, and contact data will be reported upon elsewhere.) DA, one of the two infants to die after the mother was removed, had the highest contact scores with the mother during preseparation of all subject infants. To the extent that maternal restrictiveness is associated with high contact scores, we would expect infants with restrictive mothers to score high in contact and be more affected by loss than infants of nonrestrictive mothers. However, some infants of nonrestric-

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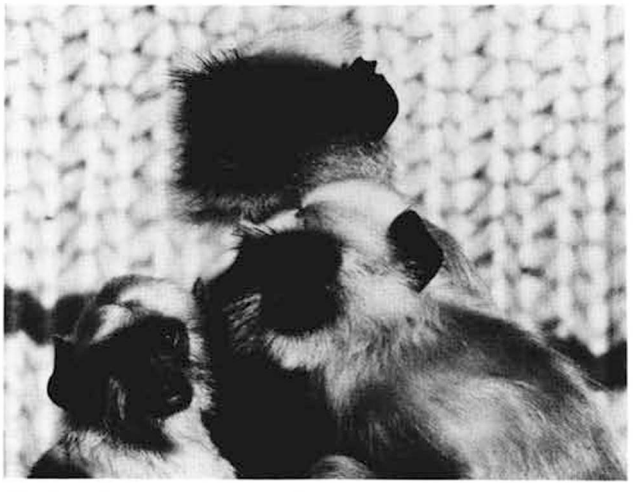


Fig. 32. The adult female LT is shown with 2 infants she accepted during their mother's absence. Both infants, one a 2-year-old male and the other an 8-month-old male, remained with LT for months.

tive mothers also had very high contact scores during preseparation but were relatively undaunted by her removal from the group.

Several females did not reject any advances by separated infants but still were not sought out or preferred as caregivers. The early experiences, from birth to the preseparation period, of each infant with all the animals in the group will be presented elsewhere. It should be noted here only that there are few correlations between relationships of an infant to any adult female other than the mother during early months of frequent infant transfers and the infant's later choice of substitute caregiver when the mother is gone.

As mentioned earlier, an important factor in the adoption of a substitute caregiver is the presence or absence or competition from other immatures for the chosen female's attentions or a place on her ventrum. Competition can come from her own immature or from other infants that have already laid claim to her attentions. The latter infants may have experienced involuntary separation from their own mother or may have left their mother because of weaning. The more persistent the infant, the greater the likelihood of a successful infant-initiated adoption. The attitude of most adult females that are of generally accepting nature appears to be to let the immatures, including their own infants, settle the

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Fig. 33. An 8-month-old male infant adopted the adult female BF although she had an 8-week-old male infant of her own. Here both infants attempt to share contact with her.

conflict. This permissiveness can sometimes have severe and life-threatening repercussions for the female's own infant if it happens to be very young.

5 infants that underwent separation had related adult females in the group that could have been approached for care, but none of these 5 infants approached or sought attention from the related females. 1 of the 5 (PH) did not gain a substitute during his mother's absence, although there were 2 related adult females in the group. He did not try to gain attention from either female.

When mothers are reintroduced in the group, if the infant is going to return to her it does so straightaway. No mother refused her infant and none searched for it. Just as the choice of a substitute caregiver was infant-initiated, so too were all of the reunions. The fact is, of 9 completed separations, 5 infants elected to remain with their substitute caregiver after the mother was returned to the group. 2 mothers (GI and WA) were removed twice from the group for

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successive infants, and each female had 1 infant return to her and 1 stay with the caregiver it had chosen in her absence.

Spontaneous adoptions, not experimentally induced, have occurred in our colony groups as the result of weaning conflict between a mother and her offspring.

Further substantiation of the potential for adoption in langurs *(P. entellus)* has been provided by field data [Boggess, 1976]. Boggess observed a social group of langurs in Nepal and was able to document an instance in which a very young infant (3 weeks old at the beginning of the transition period) was 'adopted' by an adult female whose own infant had died recently. These field data are very significant not only because they document the process of change in primary caregiver, but also because they document the fact that mothers exhibit very different degrees of interest in caring for (and, in fact, in keeping) their infants.

The experimental removals of mothers as reported here have helped demonstrate that langur infants have the ability to make strong and irreversible shifts from a primary caregiver to another adult female – and that these shifts can last for months. We do not know yet the minimum length of time the mother must be missing in order to stimulate permanent change of caregiver, but this question is being investigated.

To understand the complex factors involved in the effects of mother loss, it is necessary to look not only at the behavior and experiences of the infant but also at the temperament and experiences of the mother, and at the social context in which loss occurs.

The evolutionary implications of the behavior of accepting nonrelated adult females to infants seeking care, and of the high rate of success and permanent adoptions during separation periods will be discussed elsewhere. From the data presented here, it is clear that the langur monkey infant's ability to gain care during a period of mother loss and associated stress has great importance for the infant in terms of its increased chances of survival should the mother die. The cost to unrelated other immatures that may be displaced from their own mothers, and the cost to unrelated adult females may match the subject infant's gain, and remain to be clarified.

Summary

In order to investigate the impact of mother loss in a multiple care-taking monkey, langur monkey mothers were individually removed from their infants' natal social group for a period of 2 weeks when the infants were 6–8 months old (mean separation age 32 weeks).

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Infant responses varied in intensity and duration, but all 12 infants were agitated initially and sought for the lost mother. On day 2 of the mother's absence, all infants experienced some degree of depression of normal activities. 2 infants died within 3 days of the removal of their mothers, but all other infants undergoing completed separation recovered from the loss. Recovery rates for various behaviors differed, and not all behaviors returned to normative levels by the end of the postseparation period.

Most infants adopted substitute caregivers in their mother's absence, and a majority of infants elected to remain with the adopted female of their choice when the mother was returned to the group. Significantly, the infants initiated adoption. Females allowed or rejected infant attempts to obtain care but did not approach infants or initiate contacts. From the first day of life, langur infants demonstrate preferences for care-taking by different females and determine their caregivers and companions as soon as their motor abilities allow. In the experimental situation, the complex nexus of factors involved in infant initiative and female receptiveness warrant further investigation.

No unitary concept of protest/despair in response to mother loss seems appropriate for the langur monkey. Qualities of mothering predispose the infant to effectively seek substitute care-giving, but the availability of such care may be crucial to the degree of separation distress. Infant persistence is of no avail if the target female is not receptive. The responses of any species to mother loss would presumably depend on a number of factors involving adults as well as immatures, and the langur presents a very useful contrast to the well-known macaques. Adult langurs participate in the very different pattern of multiple care-taking, and infant langurs experience a very different early social development and are less prone to suffer severe stress upon maternal separation. It remains to elucidate the reasons for these differences, but they will surely include the multiple care-giving pattern, infant initiative and adult acceptance. With broader understanding and new models, we hope to understand what causes infant distress and depression and how they can be prevented.

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